What Does the Price of TIPS Depend on?

Research Paper

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1 Abstract

The literature has long documented a discrepancy between nominal US Treasury yields and those implied by a replicating portfolio of Treasury Inflation-Protected Securities (TIPS) and inflation swaps. This wedge is usually attributed to a liquidity premium embedded in TIPS and, in some cases, interpreted as evidence of a profitable arbitrage opportunity. We show that liquidity is only part of the story. Using a model-free decomposition of Treasury yields and inflation swap rates, we find that the observed spread depends not only on liquidity conditions but also on inflation surprises. This second effect becomes apparent after the 2022 inflation shock, when deteriorating liquidity coincided with a sharp increase in the demand for inflation protection. During this period, TIPS yields fell despite worsening liquidity, allowing the two effects to be disentangled. We further decompose nominal and real Treasury yields into expectations and risk premia associated with inflation and real rates. We find that long-term inflation expectations remained broadly anchored despite the post-COVID inflation surge, while real risk premia turned negative, consistent with investors treating TIPS as a hedge against adverse inflation outcomes. Overall, our findings suggest that the apparent arbitrage between nominal Treasuries, TIPS, and inflation swaps largely reflects compensation for liquidity and inflation-related risks rather than a persistent market inefficiency.

2 Introduction

This paper analyzes the determinants of the spread between a nominal US Treasury bond and its replicating portfolio. It also provides a model-free decomposition of nominal and real yields into components reflecting expected real rates, expected inflation, and the associated risk premia, along with an additional term capturing liquidity effects and inflation surprises.

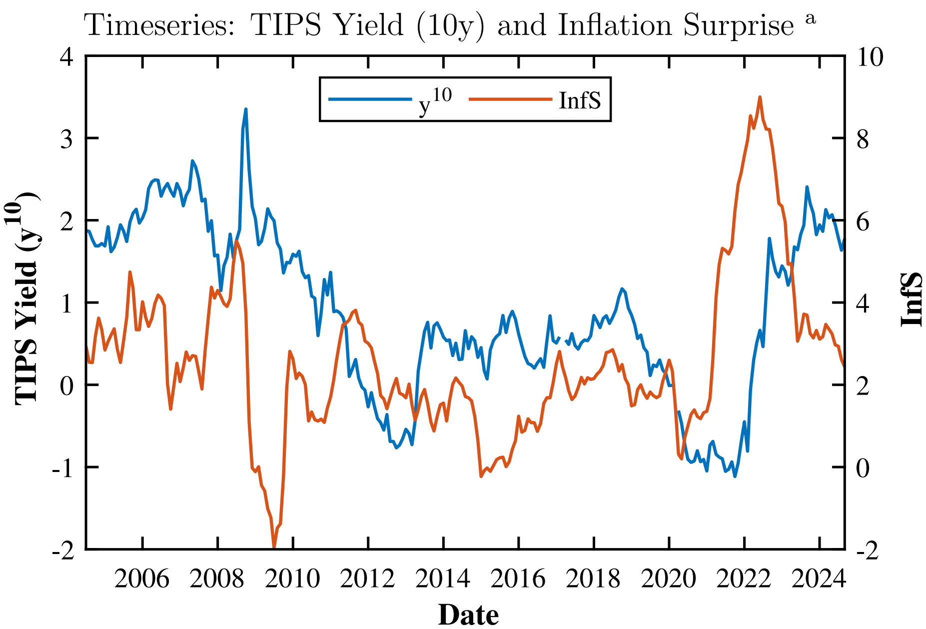
Starting with the first part of this paper, it is well known that, as documented by Fleckenstein et al. (2014), the payoffs of a nominal Treasury bond can be (almost) exactly replicated at expiry by trading in TIPS (Treasury Inflation-Protected Securities) and entering an inflation swap. However, as the literature on limits to arbitrage makes clear[1], the law of one price, dictating that the price of the nominal Treasury bond and the cost of the replicating portfolio of a TIPS and an inflation swap should be the same, holds only if investors maintain the replicating portfolio to maturity. In reality this is unlikely to be the case, especially given that these strategies are characterized by very long maturities, sometimes stretching decades into the future. If the replicating portfolio has to be unwound before maturity, there is no guarantee that the prices of its components should add up to the price of the nominal bond.

We show that this price discrepancy between the nominal Treasury bond and the replicating portfolio depends not only on liquidity, as already suggested by Abrahams et al. (2016) and D'Amico et al. (2018), but also on inflation surprises. The dependence on liquidity is easy to understand: since TIPS are known to be less liquid than nominal Treasuries, and states of poor liquidity tend to be associated with states of high market distress, hence low consumption, one can expect that TIPS should attract a positive liquidity risk premium.

[1]See Shleifer (2000) for further details.

The dependence on inflation surprises is more surprising, especially because the price of TIPS does not depend on expected inflation. What happens is that, in case of positive inflation surprises, the demand for TIPS increases pushing prices up and vice versa. Because the demand for TIPS depends on inflation surprises, one has to distinguish between situations of market distress associated with declining inflation expectations (*recessionary* conditions), and scenarios in which market distress is caused by an increase in inflation (*inflationary* conditions). Studies conducted before the 2022 inflation shock were not able to disentangle this double dependence, because the greatest liquidity shocks in the data were associated with the 2008 Great Financial Crisis during which the fall in liquidity and in inflation expectations pulled TIPS yields in the same direction. It is only with the post-2022 data that the two effects can be disentangled: despite the deterioration in market liquidity observed over this period (e.g., as proxied by VIX), TIPS yields fell significantly because of the sharp increase in inflation expectations. Although a more formal analysis is presented in Section (7), Figure (1) already hints this dependence of TIPS yields on inflation surprises.

FIGURE I



[a]This figure presents the timeseries of the 10-year TIPS real yield and of Inflation Surprise from August 2004 up to September 2024. All variables are reported in percentage. Inflation Surprise is calculated as the difference between the one-year inflation rate and the 10-year moving average. Further colour about these metrics will be provided in Sections (4) and (6).

Moving to the second part of the paper, we show how information about the Federal Reserve monetary intentions, as reflected in the *Blue Dots*, can be used to decompose nominal and real yields into expectations and risk premia associated with real rates and inflation.

The results of the proposed yield decomposition are conditional on the estimation of expected inflation. Given its pivotal role in our analysis, we estimate it using two different approaches: either econometrically or by sourcing it from the projections provided by the Federal Reserve[2]. Although estimates from the two approaches differ, especially at short maturities, we find that those discrepancies are largely *de-minimis* at long tenors. Hence, given the lower dispersion of expected inflation estimates, our

[2]Haubrich et al. (2012) provides the term structure of expected inflation by estimating a model of nominal and real bond yields using nominal Treasury yields, inflation swap rates, and inflation survey forecasts.

findings are more robust at longer tenors. Since our data include the 2022 inflation surge, this aspect is important to assess whether:

- a higher demand for TIPS as an inflation hedge in an inflationary environment has resulted in the compression of real risk premia (more expensive TIPS), and
- the inflation risk premium has risen during phases of rising inflation.

We will show from inflation projections that market participants regard US inflation as credibly anchored, consistent with the view that inflation uncertainty is currently not a material concern warranting significant risk compensation in nominal Treasury yields[3], and that the real risk premium became negative in 2022 supporting the idea that TIPS are purchased by consumers as a consumption hedge during periods of high inflation.

In sum, our findings are that: (i) TIPS yields offer a liquidity premium relative to nominal Treasury yields, (ii) inflation expectations remain anchored to the Federal Reserve target, even following the post-COVID surge in inflation dynamics, and (iii) real risk premia move inversely to inflation surprises, as TIPS represent a hedging tool during inflation surges.

This paper is organized as follows:

- Section (3) (*Literature Review*) outlines the two building blocks supporting this analysis: (i) factors affecting TIPS yields, and (ii) the existence of a potential market arbitrage among TIPS, nominal Treasuries, and inflation swaps due to the biases embedded in the BEI.

[3] This is consistent with the explicit inflation target set by the Federal Reserve in January 2012 and the view that monetary policy is committed to successfully counteracting short-term inflation volatility, as observed in Chernov and Mueller (2012) and Bauer (2018).

- Section (4) (*Data Description*) provides an overview of the data used and the corresponding sources.

- Section (5) (*Definitions and Assumptions*) introduces the variables and the assumptions employed throughout the article.

- The purpose of Section (6) (*Methodology*) is to elaborate on the model underlying the two goals of this paper. Hence, this part describes the approaches and additional inputs (e.g. liquidity proxies and measures of expected inflation) used to explain the yield difference between nominal Treasuries and TIPS, and to decompose real and nominal yields.

- Section (7) (*Results and Discussion*) substantiates the assumptions made in Section (6) and presents the results obtainted from the methodology employed.

3 Literature Review

This section highlights key references on the sources of the bias affecting the BEI, and whether documented evidence of underpriced TIPS valuations in comparison to nominal Treasuries reflects exposure to previously unaccounted factors, or if it is just a market inconsistency paving the way to a persistent arbitrage opportunity.

A TIPS is a debt security issued by the US Treasury, where the bond principal is linked to the realized changes in the non-seasonally adjusted US consumer price index for all urban consumers (CPI), published by the US Bureau of Labor Statistics (BLS). Since semi-annual interest payments are calculated on the inflation-adjusted notional, the coupons, although at a fixed rate, are harmonized for inflation as well[4]. Because TIPS offer inflation protection and are issued by the same entity as nominal Treasuries, the BEI[5] might be considered a real-time, unbiased proxy of inflation expectations by market participants.

The existence of an arbitrage among inflation swaps, nominal US Treasuries, and TIPS was initially documented by Fleckenstein et al. (2014) and further assessed in Bretscher (2015), Driessen et al. (2017), To and Tran (2019) and Kita and Tortorice (2021).

The structure of this trading strategy is relatively simple and consists of replicating the cashflows of a Treasury bond by taking both: i) a long position in a TIPS, with the same maturity as the nominal bond, and ii) the inflation-payer leg in a portfolio of inflation swaps[6]. The portfolio of inflation swaps consists of one fixed-rate receiver position for

[4]Further details about the structure of a TIPS are available in Sack and Elasser (2004).

[5]At a given maturity, the BEI is defined as the difference in yields between a nominal security and an inflation-linked bond of the same residual term, both issued by the US Treasury.

[6]This derivative product has one counterpart paying the cumulative CPI over the term of the contract at maturity against a predefined fixed rate discretely compounded, set at trade inception. The fixed rate is set such that the inflation swap has a zero value at inception. Further details on this product are available in Jarrow and Yildirim (2023).

each interest date of the TIPS, with a notional equal to the TIPS coupon for regular payment dates and the coupon plus the bond principal at maturity[7]. Since the purpose of the inflation swap portfolio is to replicate the nominal Treasury by converting all the inflation-linked cashflows into a nominal stream of payments, it follows that a positive expected return might arise from the discrepancy between the BEI and the inflation swap rate. Indeed, in the absence of other factors affecting TIPS prices, we would expect the two rates to be equal at a given tenor, since TIPS and inflation swaps are linked to the same price index. If the BEI exceeds the inflation swap rate, an investor, keeping her positions to expiry, could earn a systematic profit by shorting the nominal Treasury and taking a long position in the corresponding replication portfolio up to its maturity, and vice versa.

As it is widely acknowledged in the literature (e.g. in Abrahams et al. (2016) and D'Amico et al. (2018)), the BEI is not an unbiased real-world expectation of inflation because of several factors:

- Inflation Risk Premium

  Nominal Treasury yields include an inflation risk premium, arising from the covariance between the real stochastic discount factor (SDF) and expected inflation[8]. A positive premium exists if the covariance between those two variables is negative, and it corresponds to the investor's reward for holding a nominal security when marginal utility from real consumption is high and purchasing power is low. In this scenario, a nominal bond represents a poor consumption hedge since the real value of its cashflows is low (high inflation) at a time when the investor needs them the most (high real SDF). Although the inflation risk premium is

[7] As commented in Fleckenstein et al. (2014), the arbitrageur needs to take either a long or a short small position in Treasury STRIPS for each interest payment date to perfectly match the nominal Treasury cashflows and keep the strategy cash-neutral.

[8] See Christensen et al. (2010) and Kupfer (2019) for further details.

usually thought to be positive, it may become negative in case of demand-driven economic shocks causing real growth and inflation to move in the same direction.

- Indexation Lag

  Since the BLS releases CPI data with some delay, there is a two-and-a-half-month lag in the TIPS inflation indexation process[9]. As a result, TIPS investors are compensated for the realized price change that occurred 2.5 months before the bond issuance date but they are left exposed to the inflation risk for the same period before bond maturity. While short-term TIPS are usually quite affected by this feature[10], their associated risk premium tends to be negligible, amounting to just a few basis points as assessed by D'Amico et al. (2018).

- Deflation Option

  TIPS include a deflation floor: the principal is guaranteed at par at maturity, regardless of cumulative deflation[11]. This embedded put raises TIPS prices and lowers the corresponding yield to maturity. While typically minor, its impact can be significant during periods of elevated deflation risk, causing the BEI to exceed inflation swap rates. Furthermore, as each TIPS has its own deflation floor (based on CPI at issuance), yield dispersion can occur across securities with similar maturities but different issue dates. Thus, the BEI depends not only on residual maturity but also on the date of issuance[12].

- Liquidity

[9] See Gürkaynak et al. (2010) or D'Amico et al. (2018).

[10] As mentioned in Gürkaynak et al. (2010), prices of TIPS with a remaining tenor below 2 years are quite erratic and, therefore, usually excluded from the estimation process of the corresponding yield curve.

[11] See Haubrich et al. (2012) or D'Amico et al. (2018) for further references.

[12] Using a fitted yield curve for TIPS, as in Gürkaynak et al. (2010), should reduce this effect in comparison to employing raw bond yields. See Section (4) for further details.

> As extensively discussed in the literature[13], the TIPS market tends to be less liquid than that of nominal Treasuries. Investors may, therefore, demand a liquidity risk premium for holding TIPS. This pattern became evident following the Lehman Brothers collapse in 2008, when a sell-off of TIPS and a flight to liquidity (nominal Treasuries) caused breakeven rates to fall below plausible levels. This is striking, given that both instruments are issued by the US Treasury and share identical credit risk[14].

As part of a growing literature, the work of Abrahams et al. (2016) and D'Amico et al. (2018) highlights that the BEI derived from TIPS is affected by liquidity, hence challenging its interpretation as a direct measure of expected inflation even after adjusting for minor effects, such as the deflation option and the indexation lag. D'Amico et al. (2018) models TIPS and nominal Treasury yields in a no-arbitrage Gaussian affine term structure framework that allows for a wedge between observed TIPS yields and underlying real yields. This wedge is treated as a latent TIPS-specific factor, interpreted mainly as a liquidity premium, and is found to be sizable, particularly in the early years of the TIPS market and during the global financial crisis in 2008. Abrahams et al. (2016) adopts a similar affine term structure setup but: (i) it models liquidity using a proxy constructed from TIPS market microstructure data (e.g. relative trading volumes), and (ii) it estimates the Gaussian affine term structure model by using linear regressions. The approach applied in Abrahams et al. (2016) yields a decomposition of BEI into expected inflation, inflation risk premia, and a liquidity component, which is found to be quantitatively relevant especially during periods of market stress. Hence, liquidity risk is identified to be the risk factor compensating investors in the trading strategy outlined by Fleckenstein et al. (2014) *et al.*

[13]See Christensen et al. (2010), Haubrich et al. (2012), Abrahams et al. (2016), D'Amico et al. (2018), and Kupfer (2019).

[14]See Haubrich et al. (2012).

4 Data Description

This part lists the timeseries and sources used in this paper and answers two questions: (i) *which raw yields/rates are employed for nominal Treasuries, TIPS, and inflation swaps?*, and (ii) *what is the time frequency and horizon of the data?*.

Nominal and inflation-linked US Treasuries are modelled as discount bonds to avoid the effects of coupon payments and intermediate cashflows on the respective yields. This modelling choice facilitates a simple calculation of the BEI and mitigates bond-specific distortions over time and across tenors[15]. Furthermore, this assumption aligns with market standard in the inflation swap market, where contracts typically feature no intermediate payments. Therefore, a single payment at maturity allows for a more straightforward comparison between inflation swap rates and the BEI and avoids complexities arising from indexation lag, CPI seasonality, and publication timings.

Therefore, in line with the observations outlined in Section (3) regarding TIPS yields, as per Abrahams et al. (2016) and D'Amico et al. (2018), this paper uses zero-coupon rates for nominal and inflation-linked bonds fitted following Gürkaynak et al. (2007) and Gürkaynak et al. (2010), rather than constructing the corresponding term structures from individual securities. Consequently, potential distortions to the BEI, arising from: (i) bond-specific liquidity effects related to market microstructure[16], (ii) the embedded deflation floor, and (iii) inflation seasonality patterns, are attenuated, allowing for a clearer focus on TIPS-specific premia.

Since this analysis depends on several timeseries characterized by either a daily (bond

[15] For example, no adjustment to the BEI is needed to correct for the duration profile of a TIPS caused by a specific inflation path.

[16] As an example: (i) issue size and outstanding amount, (ii) on-the-run versus off-the-run status, (iii) dealer inventory and market-making coverage.

yields and inflation swap rates) or monthly (CPI/inflation rates) frequency[17], we clarify how those timeseries are aligned and comment on the time horizon covered in this paper. In order to align the daily market data timeseries of bond yields and inflation swap rates with CPI publication dates, the frequency of all variables is set at monthly, and each step falls on the inflation release dates. It follows that for every month of bond yield data available, this paper considers the yields observed on the inflation release date only. Additionally, it is worth noting that, although there is a release delay of approximately two weeks between the end of the reporting month and the actual CPI publication date by the BLS, our choice of date alignment places the emphasis on when the market actually receives new information and can, therefore, revise expectations and yields accordingly, rather than on the end-of-month reporting date. The time horizon covered in this analysis focuses mainly on the period between August 2007 and September 2024. Even if the start date of the timeseries at hand usually occurs before August 2007, this choice is driven by the utilisation of selected data to implement a principal component analysis supporting the construction of a liquidity factor[18]. In addition, we wanted to give time to the TIPS market to become more liquid and established.

Following the determination of the type of yields/rates used for bonds and inflation swaps and the discussion of the frequency and length of the timeseries to use, we identify the range of bond tenors assessed and the data sources used in this paper. Due to limitations in the availability of Treasury yield data, the range of tenors defining the cross-section of maturities assessed comprises the interval: $[2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 15, 20]$ years. In addition, no maturities shorter than two years are considered due to estimation challenges for TIPS yields, as highlighted in Gürkaynak et al. (2010).

[17] Further colour on the actual timeseries will be provided later in this section.
[18] Further colour on this topic will be provided in Section (6.1.1).

The data used are publicly available and retrievable via Bloomberg, the Bureau of Labor Statistics, and the Federal Reserve websites:

- The zero-coupon timeseries for nominal Treasuries and TIPS yields are daily and updated weekly on the Federal Reserve website[19].
- VIX is calculated daily by the CBOE and sourced from Bloomberg using ticker: *VIX Index*.
- The on-/off-the-run 10-year US Treasury spread and the average yield error in the US Treasury market are available daily in Bloomberg at tickers: *G0111Z 10Y BLC2 Curncy* and *GVLQUSD Index*.
- The synthetic zero-coupon inflation swap rates are daily and sourced from Bloomberg under ticker: *USSWITY*.
- CPI monthly data are available on the Bureau of Labor Statistics website[20].
- Federal Reserve estimates of expected inflation are updated monthly and are available on the website of the Federal Reserve Bank of Cleveland[21].
- The *Dot Plot* graph is released quarterly (currently in March, June, September, and December), starting in 2012[22], in the Summary of Economic Projections. This report is available on the Federal Reserve website[23].

[19]See https://www.federalreserve.gov/data.
[20]See https://www.bls.gov/cpi.
[21]See https://www.clevelandfed.org/indicators-and-data/inflation-expectations.
[22]It is worth noting that this timeseries impacts the time horizon at hand only for the full decomposition of nominal and real yields.
[23]See https://www.federalreserve.gov/monetarypolicy/fomccalendars.htm.

5 Definitions and Assumptions

Following the discussion of the data, this section is divided into two parts: the first introduces the general setup (notation and definitions) of the variables underpinning our methodology, while the second outlines the assumptions underlying the decomposition of Treasury yields and inflation swap rates.

5.1 Definitions

We provide a list of the variables used throughout the paper:

- T is the maturity of a security and $t \leq T$ a generic month.
- $\tau = T - t$ defines the residual term of a bond or the period over which inflation is projected.
- Y_t^T and y_t^T correspond to the yield to maturity of a nominal zero-coupon Treasury and of a discount TIPS at time t, respectively.
- $\pi(t)$ and $r(t)$ are the instantaneous inflation and real rates in t.
- $R(t) = \pi(t) + r(t)$ is the instantaneous nominal rate observed in t.
- $\hat{\pi}_t^T = \frac{\mathbb{E}^{\mathbb{P}}\left[\int_t^T \pi(s)ds\right]}{\tau}$ is the expected average inflation between t and T.
- $\hat{r}_t^T = \frac{\mathbb{E}^{\mathbb{P}}\left[\int_t^T r(s)ds\right]}{\tau}$ is the average short-term real rate between t and T.
- rpr_t^T and $rp\pi_t^T$ represent the risk premia for the real rate and inflation risks over τ.
- λ_t^T is the compensation in t for the risk factor associated with holding a TIPS, maturing in T, rather than the corresponding nominal discount bond.

- ISR_t^T denotes the zero-coupon inflation swap rate at time t for maturity T.

- $\mathbb{P}$ and $\mathbb{Q}$ are the historical and risk-neutral probability measures, respectively.

5.2 Assumptions

Now that we have presented the setup of the variables, we introduce the three key assumptions behind the decomposition of Treasury yields and inflation swap rates into risk premia and expectations.

The proposed decomposition relies on the following assumptions:

- TIPS yields are exposed to a latent factor (λ_t^T) that does not affect nominal Treasury yields, representing liquidity effects and inflation surprises.

- The inflation swap rate curve is an unbiased estimate of expected inflation plus the corresponding risk premium. Hence, additional factors such as balance sheet costs, exposure margining, or worsened market liquidity do not affect the mid price but only widen the bid-ask spread of inflation swap rates[24].

- Yield convexity for bond yields and inflation swap rates is assumed negligible[25].

Given the assumption of no yield convexity, we define the yield to maturity of a nominal zero-coupon Treasury as the sum of the real and inflation rates over its residual term,

[24] This assumption will be validated in Section (7.1.2) by testing whether inflation swap rates depend on market liquidity measures (e.g. VIX).

[25] A convexity adjustment for Treasury yields (TIPS and nominal) at maturities up to 10 years is marginal, as shown in Rebonato and Ronzani (2021). Similar evidence is documented by Ang et al. (2008) for inflation swap rates. Furthermore, after estimating yield volatility, the convexity contribution is assessed to be below 7 basis points for the longest bond maturity considered (20 years).

plus a risk premium compensating for exposure to inflation and real rate risks:

$$Y_t^T = \frac{\mathbb{E}^{\mathbb{P}}\left[\int_t^T r(s)ds\right]}{\tau} + \frac{\mathbb{E}^{\mathbb{P}}\left[\int_t^T \pi(s)ds\right]}{\tau} + rpr_t^T + rp\pi_t^T \tag{1}$$

At the same time, the yield to maturity of a discount TIPS is defined as the sum of the real rate and its corresponding risk premium over τ, plus the TIPS-specific compensation for latent risks:

$$y_t^T = \frac{\mathbb{E}^{\mathbb{P}}\left[\int_t^T r(s)ds\right]}{\tau} + rpr_t^T + \lambda_t^T \tag{2}$$

Making use of the second assumption, inflation swap rates are defined as:

$$\begin{aligned} ISR_t^T &= \frac{\mathbb{E}^{\mathbb{P}}\left[\int_t^T \pi(s)ds\right]}{\tau} + rp\pi_t^T \\ &= \hat{\pi}_t^T + rp\pi_t^T \end{aligned} \tag{3}$$

It follows that we can use Expressions (1) and (2) to define the BEI:

$$\begin{aligned} BEI_t^T &= Y_t^T - y_t^T \\ &= \frac{\mathbb{E}^{\mathbb{P}}\left[\int_t^T \pi(s)ds\right]}{\tau} + rp\pi_t^T - \lambda_t^T \end{aligned} \tag{4}$$

Hence, Formula (4) shows that BEI_t^T is an unbiased measure of the expected inflation over τ only when $rp\pi_t^T = \lambda_t^T = 0$.

6 Methodology

In line with the two questions of this paper: (i) *do TIPS yields embed a latent component?* and (ii) *how do we decompose nominal (and real) yields?*, this section is split in two parts. The first part introduces the methodology used to identify whether liquidity and inflation surprises impact TIPS yields in *Factors Affecting TIPS Yields*, and the second part presents the method underpinning the implementation of the actual decomposition of Treasury yields and inflation swap rates into risk premia and expectations in *Decomposition of Nominal (and Real) Yields*.

6.1 Factors Affecting TIPS Yields

Making use of Section (5.2), which identifies the components underpinning Y_t^T, y_t^T and ISR_t^T in Equations (1), (2) and (3), we can isolate the TIPS-specific compensation for unaccounted risks as the difference between the inflation swap rate in Equation (3) and the corresponding BEI as per Formula (4):

$$\lambda_t^T = ISR_t^T - BEI_t^T \tag{5}$$

It is worth noting that, so far, we have relied only on readily observable quantities to extract λ_t^T, without making use of any estimates of expected inflation or risk premia beyond the assumptions so far presented. Clearly, the result in Formula (5) depends on the validity of the second assumption presented in Section (5.2).

Following the introduction of a method to calculate λ_t^T as per Formula (5), we focus on whether this metric represents compensation for liquidity risk and exposure to inflation surprises. Since liquidity is latent and can be captured only by proxies, validating whether λ_t^T embeds a liquidity premium involves: (i) defining proper liquidity proxies and (ii) testing the appropriateness of the assumption related to the inflation swap rate curve. Therefore, we outline in Section (6.1.1) how liquidity is proxied and use these metrics to substantiate our assumptions in Section (7.1.2). In addition, we need to ensure that λ_t^T is properly identified as compensation for liquidity risk. To this end, we employ an estimate of expected inflation[26] to independently extract λ_t^T from BEI_t^T without relying on ISR_t^T. This quantity is defined as follows:

$$\begin{aligned} BEI_t^T - \hat{\pi}_t^T &= rp\pi_t^T - \lambda_t^T \\ &= \hat{\gamma}_t^T \end{aligned} \tag{6}$$

At the same time, we regress λ_t^T on a proxy for inflation surprises in order to capture whether any residual component of the yield difference between nominal Treasury bonds and TIPS depends on inflation shocks[27].

In sum, we separately regress $\hat{\gamma}_t^T$, λ_t^T and ISR_t^T on liquidity and λ_t^T on a proxy for inflation surprises to assess whether the conjecture that TIPS yields embed a latent factor is valid:

$$\begin{aligned} \hat{\gamma}_t^T &= \alpha_{\hat{\gamma}}^T + \beta_{\hat{\gamma}}^T PCA_t + \epsilon_{t,\hat{\gamma}}^T \\ \lambda_t^T &= \alpha_{\lambda}^T + \beta_{\lambda}^T PCA_t + \epsilon_{t,\lambda}^T \\ ISR_t^T &= \alpha_{ISR}^T + \beta_{ISR}^T PCA_t + \epsilon_{t,ISR}^T \\ \lambda_t^T &= \alpha_{InfS}^T + \beta_{InfS}^T InfS_t + \epsilon_{t,InfS}^T \end{aligned} \tag{7}$$

[26]See Section (6.2.1) for further details.

[27]It is worth noting that we are trying to capture two distinct effects within λ: liquidity and inflation surprises. Therefore, since we are using two separate linear regressions, we will expand on the correlation between the two independent variables in Section (7.1), observing that the corresponding cross-correlation effects are rather small and thus can be ignored.

The time step and horizon dimensions are defined as per Section (4). In addition to the definitions presented in Section (5.1), PCA_t is the first liquidity proxy, interchangeable with ${VIX_t}^{28}$, and $InfS_t$ represents a proxy for the inflation surprise observed at time t. This measure corresponds to the difference between actual yearly inflation and a 10-year moving average of realized 12-month inflation, such that:

$$InfS_t = \frac{CPI_t - CPI_{t-12}}{CPI_{t-12}} - \frac{1}{120} \sum_{j=t-119}^{t} \frac{CPI_j - CPI_{j-12}}{CPI_{j-12}}$$

The realised inflation rate is calculated using the seasonally adjusted CPI and, since the CPI timeseries starts well before August 2007, the moving average is computed over the full 120-month window from $t - 119$ to t, without any truncation.

These regressions follow a standard OLS framework, further background around additional econometric tools supporting the qualitative results of this paper will be presented in Appendix A.

6.1.1 Creating Liquidity Proxies

The separate regressions of $\hat{\gamma}_t^T$ and λ_t^T on liquidity are necessary to assess whether TIPS yields compensate investors for liquidity risk. Since liquidity is a latent factor and not directly observable, the purpose of this section is to introduce market variables that can serve as proxies for liquidity risk.

As noted in Nagel (2012), Drechsler et al. (2018), and Drechsler et al. (2020), liquidity risk is typically proxied by market-based measures of volatility, such as MOVE[29] and,

[28] Both metrics will be discussed in Section (6.1.1).

[29] The MOVE index, constructed by ICE BofA, measures the implied volatility of US Treasury yields across multiple maturities. It is computed as a weighted average of implied volatilities from options on Treasury securities and provides a forward-looking gauge of interest rate uncertainty.

in particular, VIX[30]. This reflects the fact that liquidity provision entails exposure to short volatility risk: liquidity providers absorb order imbalances in normal times, hence earning a premium, but incur losses during periods of market stress, when volatility spikes.

VIX is the main candidate to proxy for liquidity risk; however, it may capture more than pure liquidity effects, as it is influenced by broader market dynamics. As a result, we may observe co-movements between our dependent variables and VIX during periods of market distress, driven by revisions in risk premia, expectations, and broader market risk aversion. Therefore, we employ a more flexible measure of liquidity, similar to the measure presented in Rebonato and Sherwin (2020), extracted through principal component analysis applied to the timeseries of: (i) VIX, (ii) the on-/off-the-run 10-year US Treasury spread, (iii) the average yield error between the observed Treasury yield curve and its fitted counterpart, and (iv) MOVE[31]. In line with the approach adopted by Rebonato and Sherwin (2020), all variables are first standardized: demeaned and scaled by their standard deviation so that each timeseries has mean zero and unit variance before applying the principal component analysis. Since the additional metrics should be more direct proxies for liquidity, we expect the first principal component (PCA) to isolate a common liquidity factor shared across the four measures, while minimizing its covariance with other risk premia.

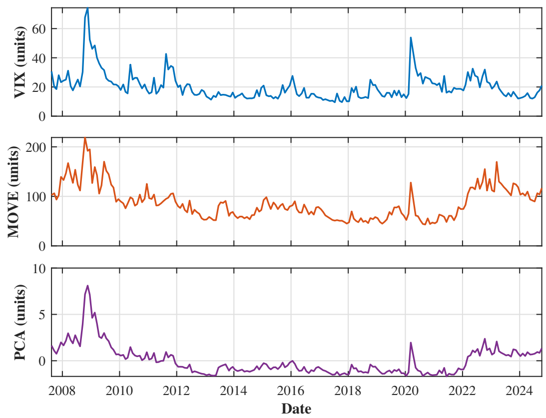
We compare the timeseries of VIX, MOVE, and PCA in Figure (2) and report both the principal component decomposition of the variance-covariance matrix of the four input variables in Table (1) and the corresponding pairwise correlations in Table (2).

[30]The VIX index, computed by the Chicago Board Options Exchange, measures the market-implied volatility of the S&P 500 over a 30-day horizon. It is derived from the prices of a wide range of out-of-the-money S&P 500 index options and is commonly interpreted as a forward-looking measure of market uncertainty and risk aversion.

[31]This paper uses the Bloomberg version of the Hu et al. (2013) liquidity index.

FIGURE II

Timeseries: VIX, MOVE and PCA[b]



[b]This figure presents the timeseries of VIX, MOVE, and the first principal component (PCA) over the sample period from August 2007 to September 2024. The first principal component is extracted from the variance–covariance matrix of: (i) VIX, (ii) the on-/off-the-run 10-year US Treasury spread, (iii) MOVE, and (iv) the average yield error between the observed Treasury yield curve and its fitted counterpart. All variables are reported in standardized units.

TABLE I

Principal Component Analysis: Eigenvalues and Explained Variance[c]

Principal Component	Eigenvalue	Explained Variance (%)
PC1	2.5640	64.1009
PC2	0.9915	24.7863
PC3	0.2758	6.8954
PC4	0.1687	4.2175

[c]This table reports the eigenvalues and corresponding explained variances of the variance–covariance matrix of: (i) VIX, (ii) the on-/off-the-run 10-year US Treasury spread, (iii) MOVE, and (iv) the average yield error between the observed Treasury yield curve and its fitted counterpart, across the four principal components. The principal component analysis covers the sample period from August 2007 to September 2024. Eigenvalues are reported in standardized units and explained variance in percentages.

TABLE II

Correlation Matrix: PCA and Input Variables[d]

Variable	Corr(PCA, Variable)
BBG_Liquidity	0.8870
VIX	0.7663
MOVE	0.9269
On_Off_Run_ZCB10Y	0.5753

[d]This table reports the correlations between the first principal component (PCA) and the four input variables: (i) VIX, (ii) the on-/off-the-run 10-year US Treasury spread, (iii) MOVE, and (iv) the average yield error between the observed Treasury yield curve and its fitted counterpart, over the sample period from August 2007 to September 2024.

Since we observe from Table (2) that the correlation between VIX and PCA is below 80%, we present results in Section (7) for both PCA and VIX as distinct liquidity proxies as a robustness check. Although PCA represents our preferred measure of liquidity, this exercise serves to reassure the reader that our results are not driven by an arbitrary choice of proxy.

6.2 Decomposition of Nominal (and Real) Yields

Now that we have provided a framework to test λ_t^T, we focus on the decomposition of nominal and real yields into risk premia and expectations. To this end, we present a linear system for y_t^T, Y_t^T, and ISR_t^T and elaborate on the additional inputs needed to close it: (i) measures of expected inflation in Section (6.2.1) and (ii) expectations of nominal yields in Section (6.2.2).

Putting together the results in Equations (1), (2), (3), and (6), we obtain a system of four equations characterized by five *unknown* variables and four observable quantities[32]:

$$\begin{cases} Y_t^T & = \hat{\pi}_t^T + \hat{r}_t^T + rpr_t^T + rp\pi_t^T \\ y_t^T & = \hat{r}_t^T + rpr_t^T + \lambda_t^T \\ \hat{\gamma}_t^T & = rp\pi_t^T - \lambda_t^T \\ ISR_t^T & = \hat{\pi}_t^T + rp\pi_t^T \end{cases} \tag{8}$$

To close the system, we need to determine a measure of inflation expectations. Indeed, the estimate of expected inflation: $\hat{\pi}_t^T$ represents the metric that enables the cross-sectional and timeseries identification of Formula (8) and, specifically, the determination of both the inflation risk premium ($rp\pi_t^T$) and the TIPS-specific compensation (λ_t^T), according to the expressions for ISR_t^T and $\hat{\gamma}_t^T$. At the same time, the premium for real rate risk remains undetermined. Therefore, while still relying on the assumption of no yield convexity, we apply a $\mathbb{P}$-measure expectation of nominal yields such that[33]:

$$\begin{aligned} \hat{R}_t^T &= \frac{\mathbb{E}_t^{\mathbb{P}}\left[\int_t^T R(s)ds\right]}{\tau} \\ &= \frac{\mathbb{E}_t^{\mathbb{P}}\left[\int_t^T r(s) + \pi(s)\right]}{\tau} \\ &= \hat{r}_t^T + \hat{\pi}_t^T \end{aligned} \tag{9}$$

By employing this metric and the estimate of the inflation risk premium embedded within ISR_t^T, we can extract the real risk premium (rpr_t^T) from the definition of Y_t^T and identify all the components of Equation (8).

[32]The observable quantities are on the left-hand side of the system, while the *unknown* variables are on the right-hand side.

[33]This metric is obtained from the *Blue Dots*. The definition and framework for this measure are provided in Section (6.2.2).

6.2.1 Measures of Expected Inflation

Although average expected inflation ($\hat{\pi}_t^T$) is not needed to determine λ_t^T, this variable is pivotal to: (i) ensure the validity of the second assumption introduced in the second part of Section (5), and (ii) to support the identification of the linear system presented in Equation (8). Therefore, the objective of this part is to describe how inflation expectations are modelled.

Since $\hat{\pi}_t^T$ is not directly observable and therefore must be estimated, we employ two different metrics:

- Federal Reserve estimates, and
- Econometric estimates based on an AR(1) model at a monthly frequency applied to the twelve-month trailing inflation rate[34].

For completeness, we use seasonally-adjusted CPI to remove the effects of predictable seasonal patterns for the AR(1) estimation, even though non-seasonally adjusted CPI is consistent with the inflation indexation process of the cashflows of both TIPS and inflation swaps. In fact, as noted in Haubrich et al. (2012), different specifications of the CPI process due to seasonality are unlikely to have a material impact on TIPS yields and swap rates, especially for medium/long-term forecasts. Therefore, we do not expect this modelling choice to significantly affect the reported results.

The decision to use two distinct measures of inflation is motivated by the need to ensure

[34]Expected average inflation is obtained by integrating the monthly projections over the corresponding forecast horizon (e.g. two years and longer maturities). In addition, it is worth noting that, although the decision to use an AR(1) model may seem arbitrary, it is justified by the nature of the underlying variable. Indeed, this monthly process is applied to an already persistent measure of inflation, namely the twelve-month trailing rate. Hence, adding further lags would result in a less parsimonious specification without improving the model's fit, given the high persistence of the underlying variable.

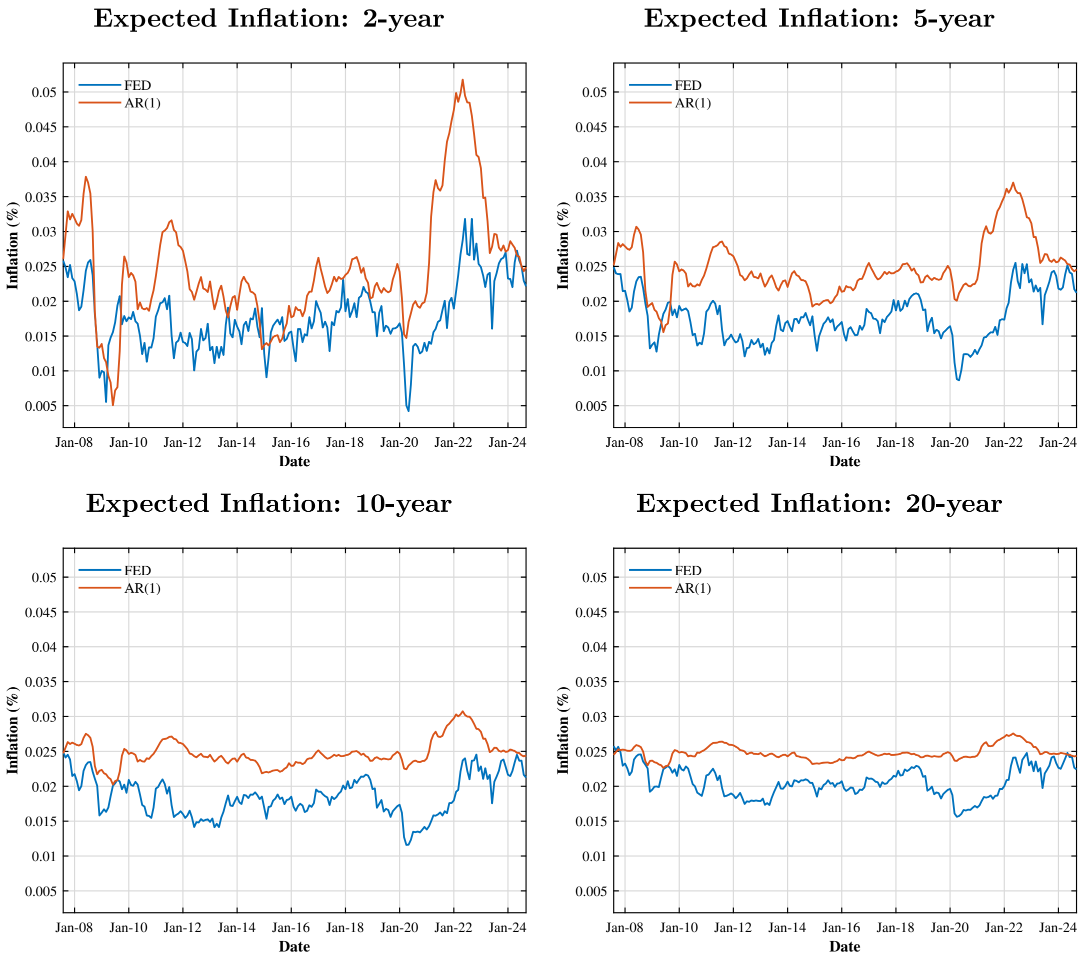
the robustness of our findings. In particular, both measures are employed in Section (7) to validate the interpretation of λ_t^T and the magnitude of the various components of nominal yields.

The respective metrics are presented in Figure (3) for visual inspection: Federal Reserve estimates in blue and AR(1)-based estimates in red. We observe that long-term inflation expectations are relatively consistent across measures. Indeed, the long-term mean reversion level hovers around 2–2.5% for both timeseries.

Nevertheless, it is worth noting that long-term Federal Reserve estimates tend to be structurally lower than the econometric measure and exhibit material volatility. Furthermore, greater dispersion in expectations is observed at shorter tenors (e.g. two years). This feature can be interpreted as reflecting differences in mean-reversion speeds of the respective inflation processes: Federal Reserve estimates appear to converge more rapidly to the target level and, thus, the corresponding stochastic process is characterized by a higher speed of mean reversion.

FIGURE III

Expected Average Inflation: AR(1) vs FED[e]



[e]This figure presents four panels showing the timeseries of expected average inflation using both Federal Reserve estimates (in blue) and AR(1)-based estimates (in red). The y-axis reports all values in percent, while the x-axis displays the corresponding dates. The sample period spans from August 2007 to September 2024.

Since the two measures of expected average inflation exhibit similar trends, we expect our results to be robust across both specifications.

6.2.2 Creating $\mathbb{P}$-Measure Expectations of Nominal Yields

To complete the framework underpinning the proxies and estimates used in this paper to decompose nominal and real yields into risk premia and expectations, we describe how the $\mathbb{P}$-measure expectation of nominal yields is specified. The definition of $\hat{R}_t^T$ is based on: (i) the assumption of no yield convexity, and (ii) the implementation of a doubly mean-reverting Vasicek (DMRV) model, based on Federal Reserve expectations of the future federal funds rate[35], to describe the path of the instantaneous nominal rate.

The decision to use a DMRV model is supported by its parsimonious parameterisation, as highlighted in Rebonato and Ronzani (2020), which facilitates modelling the levels of risk premia and expectations more effectively than other popular statistical return-predicting factors, such as those in Cochrane and Piazzesi (2005) and Cieslak and Povala (2015).

The application of the DMRV model is achieved using the *Dot Plot* chart available in the Summary of Economic Projections, whose quarterly release by the Federal Reserve began in January 2012. The *Dot Plot* chart provides individual FOMC members' anonymous and voluntary projections for future policy rates at year-end, at the end of the following year, at the end of the second year[36], and in the long run.

The expected instantaneous nominal rate at a generic instant is set as follows:

$$\begin{aligned}
\mathbb{E}_t^{\mathbb{P}}\left[R(s)\right] &= \mathbb{E}_t^{\mathbb{P}}\left[r(s)+\pi(s)\right] \\
&= R(t)e^{-k_r(s-t)} + \frac{k_r\left(\theta_t - r_\infty\right)}{k_r - k_\theta}\left(e^{-k_\theta(s-t)} - e^{-k_r(s-t)}\right) \\
&+ r_\infty\left(1 - e^{-k_r(s-t)}\right)
\end{aligned} \tag{10}$$

where:

[35]Further details are available in Rebonato (2018).

[36]This analysis excludes the third year-end projections reported in the *Dot Plot* chart for the last two quarters of each reporting year to avoid overidentifying the parameters in Formula (10).

- $R(t)$ is the current federal funds rate observed on the release date of the corresponding Summary of Economic Projections.

- r_∞ represents the average of the long-run *Blue Dots* provided in the Summary of Economic Projections.

- The respective averages of the data points in the Summary of Economic Projections at $s - t =$ year-end, end of the following year, and end of the second year are used as $\mathbb{E}_t^{\mathbb{P}}[R(s)]$ to calibrate k_r, θ_t, and k_θ on a quarterly basis.

- k_r and k_θ are the reversion speeds of the instantaneous nominal rate $(R(t))$ to the local target rate (θ_t) and of θ_t to the ergodic rate (r_∞).

- $T \geq s \geq t$.

Since the instantaneous nominal rate curve is known, $\hat{r}_t^T$ can be determined as per below:

$$\hat{r}_t^T = \frac{\mathbb{E}_t^{\mathbb{P}}\left[\int_t^T R(s)ds\right]}{\tau} - \hat{\pi}_t^T \tag{11}$$

As previously mentioned, this result enables the full identification of Equation (8) and the determination of the risk premium associated with the real rate (rpr_t^T).

7 Results and Discussion

Consistently with the two main goals of the paper, this section is accordingly divided into two parts. First, in *Factors Affecting TIPS Yields*, it clarifies whether TIPS yields embed a component for liquidity and inflation surprises. Second, in *Decomposition of Nominal (and Real) Yields*, it provides both a timeseries and a cross-sectional decomposition of TIPS and nominal Treasury yields.

7.1 Factors Affecting TIPS Yields

Here, we answer the first question of the paper: *does λ_t^T depend on liquidity and inflation surprises?* On one hand, answering this question involves testing the hypotheses made in Section (5.2): (i) λ_t^T represents compensation for liquidity risk, and (ii) ISR_t^T does not depend on liquidity. Since liquidity is latent, we rely on the two proxies outlined in Section (6.1.1). On the other hand, we will need to confirm whether any dependency of λ_t^T on $InfS_t$, making sure that cross-correlation effects between $InfS_t$ and our proxies for liquidity are rather small.

The regression analysis presented here relies on the OLS framework introduced in Section (6.1). Further results, which account for cross-sectional correlation in the structure of the residuals, are presented in Appendix A. Despite the application of a more refined econometric tool, no change to the qualitative conclusions outlined in this discussion has been observed.

7.1.1 What does λ represent?

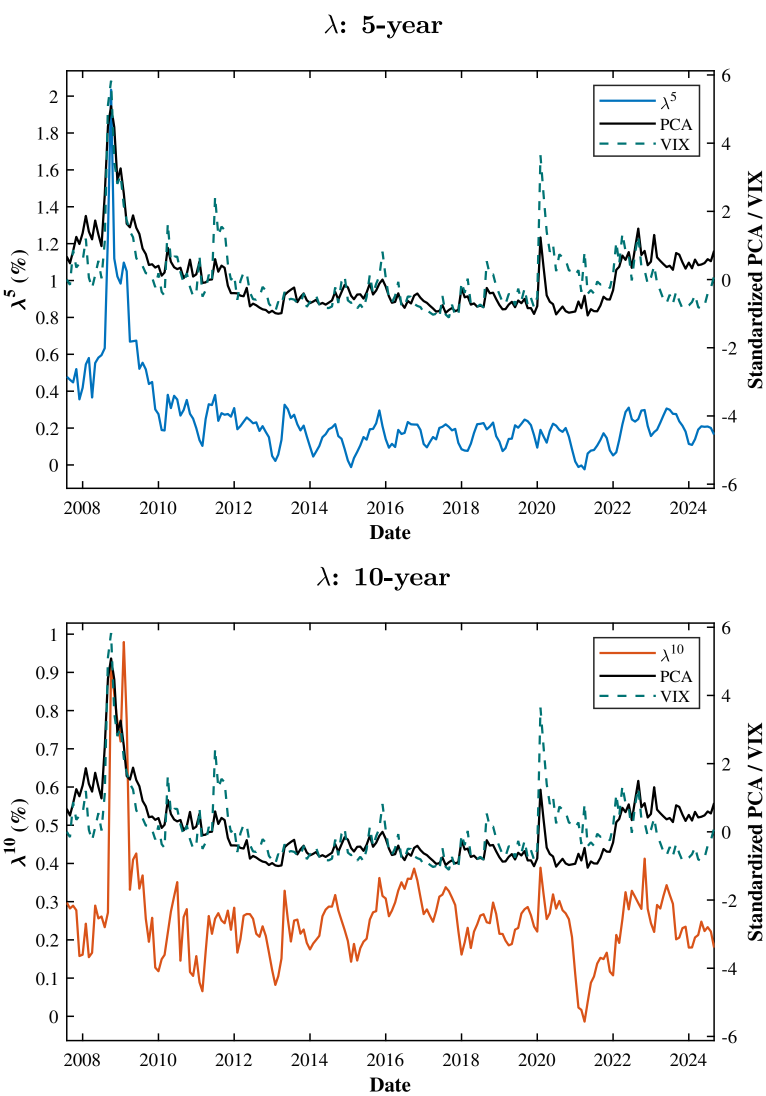
This section examines the two effects reflected by λ_t^T: (i) liquidity and (ii) inflation surprises. The first part focuses on liquidity and the two corresponding proxies, while the second looks at the impact from inflation surprises.

Assuming the validity of the *ISR/Liquidity* assumption[37] and given TIPS and nominal Treasury yields, we can isolate $\lambda_t^T = ISR_t^T - BEI_t^T$, as per Equation (5), and test whether this metric depends on: (i) PCA_t and (ii) VIX_t.

The first step in advancing our analysis is a qualitative description of λ_t^T against those variables: *are these metrics correlated?* The timeseries are reported in Figure (4):

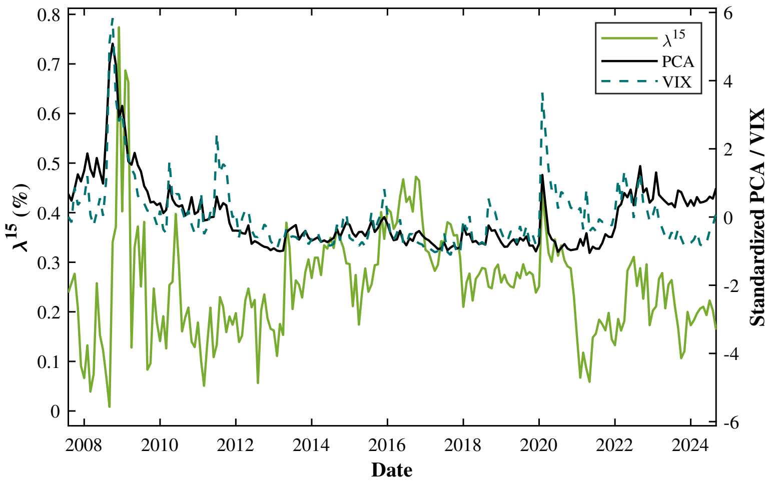
FIGURE IV

Lambda[f]

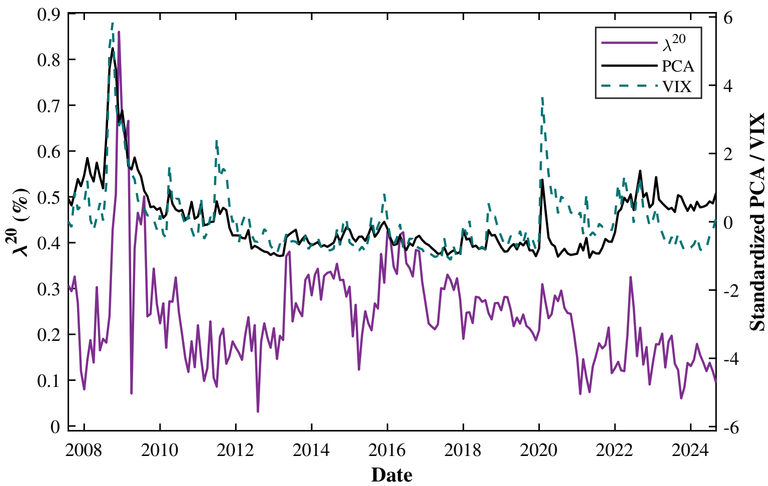


[37] This assumption will be validated in the next section.

λ**: 15-year**



λ**: 20-year**



[f]This figure presents four panels showing the timeseries of λ_t^T for $T = 5, 10, 15, 20$ years, in comparison with VIX_t and PCA_t. The y-axis reports λ_t^T in percent and VIX and the first principal component (PCA) in standardized units (each observation is demeaned and divided by the corresponding standard deviation calculated over the period from August 2007 and September 2024), while the x-axis displays the corresponding dates. The first principal component is extracted from the variance-covariance matrix of: (i) VIX, (ii) the on-/off-the-run 10-year US Treasury spread, (iii) MOVE, and (iv) the average yield error between the observed Treasury yield curve and its fitted counterpart. The sample period spans from August 2007 to September 2024.

Following this anecdotal evidence hinting at co-movements between λ_t^T and liquidity, as observed during the peaks of the Global Financial Crisis (2008–2009), the COVID-19 outbreak (2020), and the post-COVID inflation surge (2022–2023), we regress λ_t^T on

PCA_t and VIX_t, respectively, to measure the strength of this relationship.

As shown in Tables (3) and (4), the following considerations are worth noting:

- There is a consistent positive relationship between λ and PCA (or VIX); hence, irrespective of the liquidity proxy, as suggested by the R^2 values and the coefficient t-statistics across the various tenors.

- Although, on a line-by-line basis, all $\hat{\beta}^T_{PCA}$ (or $\hat{\beta}^T_{VIX}$) are statistically different from zero, it emerges that the size of the coefficients and the explained variance of the corresponding λ fade as the tenor of the TIPS increases. Subject to further research, which could be addressed in future extensions of this paper, this phenomenon may be reconciled by the type of investors holding longer-dated bonds. Indeed, Fleckenstein et al. (2014), Bretscher (2015), and To and Tran (2019) hypothesize that these effects could be caused by the market segmentation of TIPS and nominal Treasuries.

TABLE III

λ and PCA: Regression Results[g]

Tenor	$\hat{\alpha}^T$	$\hat{\beta}^T_{PCA}$	SE($\hat{\alpha}^T$)	SE($\hat{\beta}^T_{PCA}$)	$t(\alpha)$	$p(\alpha)$	$t(\beta)$	$p(\beta)$	R^2
$\hat{\lambda}^{2Y}$	0.2464	0.1053	0.0126	0.0079	19.549	0.0000	13.349	0.0000	0.466
$\hat{\lambda}^{3Y}$	0.2580	0.1299	0.0107	0.0067	24.018	0.0000	19.311	0.0000	0.646
$\hat{\lambda}^{4Y}$	0.2594	0.1352	0.0100	0.0063	25.918	0.0000	21.574	0.0000	0.695
$\hat{\lambda}^{5Y}$	0.2566	0.1232	0.0088	0.0055	29.126	0.0000	22.331	0.0000	0.710
$\hat{\lambda}^{6Y}$	0.2514	0.1032	0.0079	0.0049	31.949	0.0000	20.949	0.0000	0.683
$\hat{\lambda}^{7Y}$	0.2502	0.0904	0.0075	0.0047	33.414	0.0000	19.290	0.0000	0.646
$\hat{\lambda}^{8Y}$	0.2463	0.0710	0.0071	0.0044	34.725	0.0000	15.976	0.0000	0.556
$\hat{\lambda}^{9Y}$	0.2478	0.0556	0.0070	0.0044	35.192	0.0000	12.608	0.0000	0.438
$\hat{\lambda}^{10Y}$	0.2556	0.0439	0.0074	0.0046	34.626	0.0000	9.503	0.0000	0.307
$\hat{\lambda}^{12Y}$	0.2533	0.0142	0.0074	0.0046	34.161	0.0000	3.065	0.0025	0.044
$\hat{\lambda}^{20Y}$	0.2403	0.0158	0.0076	0.0048	31.665	0.0000	3.316	0.0011	0.051

[g]This table reports OLS estimates for $\hat{\boldsymbol{\beta}}_{\boldsymbol{\lambda}}$ and line-by-line inferential statistics such as R^2, t-stats and p-values using the first principal component (PCA_t) as proxy for liquidity. The first principal component is extracted from the variance-covariance matrix of: (i) VIX, (ii) the on-/off-the-run 10-year US Treasury spread, (iii) MOVE, and (iv) the average yield error between the observed Treasury yield curve and its fitted counterpart. The sample period spans from August 2007 to September 2024. T-stats and R^2 are rounded to the third decimal place, while all other metrics to the fourth digit.

TABLE IV

λ and VIX: Regression Results[h]

Tenor	$\hat{\alpha}^T$	$\hat{\beta}^T_{VIX}$	SE($\hat{\alpha}^T$)	SE($\hat{\beta}^T_{VIX}$)	$t(\alpha)$	$p(\alpha)$	$t(\beta)$	$p(\beta)$	R^2
$\hat{\lambda}^{2Y}$	-0.0684	0.0158	0.0334	0.0015	-2.046	0.0421	10.336	0.0000	0.344
$\hat{\lambda}^{3Y}$	-0.1296	0.0194	0.0313	0.0014	-4.138	0.0001	13.590	0.0000	0.475
$\hat{\lambda}^{4Y}$	-0.1532	0.0207	0.0296	0.0013	-5.177	0.0000	15.310	0.0000	0.535
$\hat{\lambda}^{5Y}$	-0.1159	0.0187	0.0267	0.0012	-4.349	0.0000	15.345	0.0000	0.536
$\hat{\lambda}^{6Y}$	-0.0584	0.0155	0.0234	0.0011	-2.490	0.0136	14.505	0.0000	0.508
$\hat{\lambda}^{7Y}$	-0.0287	0.0140	0.0211	0.0010	-1.357	0.1764	14.488	0.0000	0.507
$\hat{\lambda}^{8Y}$	0.0244	0.0111	0.0189	0.0009	1.293	0.1977	12.889	0.0000	0.449
$\hat{\lambda}^{9Y}$	0.0681	0.0090	0.0177	0.0008	3.843	0.0002	11.136	0.0000	0.378
$\hat{\lambda}^{10Y}$	0.1080	0.0074	0.0179	0.0008	6.028	0.0000	9.043	0.0000	0.286
$\hat{\lambda}^{12Y}$	0.1949	0.0029	0.0176	0.0008	11.089	0.0000	3.648	0.0003	0.061
$\hat{\lambda}^{20Y}$	0.1776	0.0031	0.0180	0.0008	9.862	0.0000	3.831	0.0002	0.067

[h]This table reports OLS estimates for $\hat{\boldsymbol{\beta}}_{\boldsymbol{\lambda}}$ and line-by-line inferential statistics such as R^2, t-stats and p-values using VIX_t as proxy for liquidity. The sample period spans from August 2007 to September 2024. T-stats and R^2 are rounded to the third decimal place, while all other metrics to the fourth digit.

Now that the relationship between λ_t^T and liquidity has been assessed, we apply the same method focusing on the dependency of λ_t^T on $InfS_t$. This is done by running the fourth regression in Formula (7); the results are presented below in Table (5).

It is worth noting that the observed dependence on inflation surprises is much stronger than what shown for the two proxies for liquidity at long maturities[38]. As previously hinted by the comments for Tables (3) and (4), this phenomenon could be explained by the market segmentation of investors by tenor. For instance, long-dated TIPS are mainly held by strategic players (e.g. lifers), who are less impacted by changes in liquidity but react more significantly to inflation shocks. TIPS are bid in an inflationary environment, compressing λ_t^T, as indicated by the corresponding negative $\hat{\beta}$, while short-dated bonds are kept by traders (e.g. banks or funds), which are more exposed to fluctuations in market liquidity.

TABLE V

λ and InfS: Regression Results[i]

Tenor	$\hat{\alpha}$	$\hat{\beta}_{InfS}$	SE($\hat{\alpha}$)	SE($\hat{\beta}_{InfS}$)	$t(\alpha)$	$p(\alpha)$	$t(\beta)$	$p(\beta)$	R^2
$\hat{\lambda}^{2Y}$	0.2737	-1.1211	0.0272	0.8385	10.057	0.0000	-1.337	0.1827	0.009
$\hat{\lambda}^{3Y}$	0.2862	-1.1678	0.0285	0.8780	10.045	0.0000	-1.330	0.1850	0.009
$\hat{\lambda}^{4Y}$	0.2843	-1.0350	0.0286	0.8821	9.930	0.0000	-1.173	0.2420	0.007
$\hat{\lambda}^{5Y}$	0.2717	-0.6400	0.0259	0.7970	10.503	0.0000	-0.803	0.4229	0.003
$\hat{\lambda}^{6Y}$	0.2668	-0.6487	0.0221	0.6804	12.084	0.0000	-0.953	0.3415	0.004
$\hat{\lambda}^{7Y}$	0.2656	-0.6442	0.0199	0.6125	13.360	0.0000	-1.052	0.2942	0.005
$\hat{\lambda}^{8Y}$	0.2645	-0.7465	0.0168	0.5169	15.765	0.0000	-1.444	0.1502	0.010
$\hat{\lambda}^{9Y}$	0.2687	-0.8509	0.0148	0.4545	18.212	0.0000	-1.872	0.0626	0.017
$\hat{\lambda}^{10Y}$	0.2843	-1.1575	0.0138	0.4251	20.605	0.0000	-2.723	0.0070	0.035
$\hat{\lambda}^{12Y}$	0.2883	-1.3991	0.0116	0.3569	24.889	0.0000	-3.920	0.0001	0.071
$\hat{\lambda}^{20Y}$	0.3016	-2.4476	0.0110	0.3392	27.400	0.0000	-7.216	0.0000	0.205

[i]This table reports OLS estimates for $\hat{\boldsymbol{\beta}}_{\boldsymbol{\lambda}}$ and line-by-line inferential statistics such as R^2, t-stats and p-values using $InfS_t$ as independent variable to capture inflation shocks. The sample period spans from August 2007 to September 2024. T-stats and R^2 are rounded to the third decimal place, while all other metrics to the fourth digit.

[38]Additionally, we can conclude that the liquidity proxies and the inflation surprise variable are poorly correlated since they are explaining the variance of λ at opposite tenors: short for liquidity vs long for inflation surprise.

7.1.2 Validating the *ISR/Liquidity* Assumption

The results produced so far highlight that λ_t^T depends on liquidity (and inflation surprises to a lesser extent) and that this relationship is robust to the alternative proxies we introduced. However, the *ISR/Liquidity* assumption, previously outlined in Section (5.2), has not been validated yet.

The identification of Equation (5) relies on the assumption that inflation swap rates do not embed compensation for liquidity risk. Under this assumption, Equation (3) holds, and any liquidity risk premium demanded by dealers to trade inflation swaps is incorporated into the bid–ask spread, leaving the mid-rate as an unbiased estimate of expected average inflation plus the inflation risk premium.

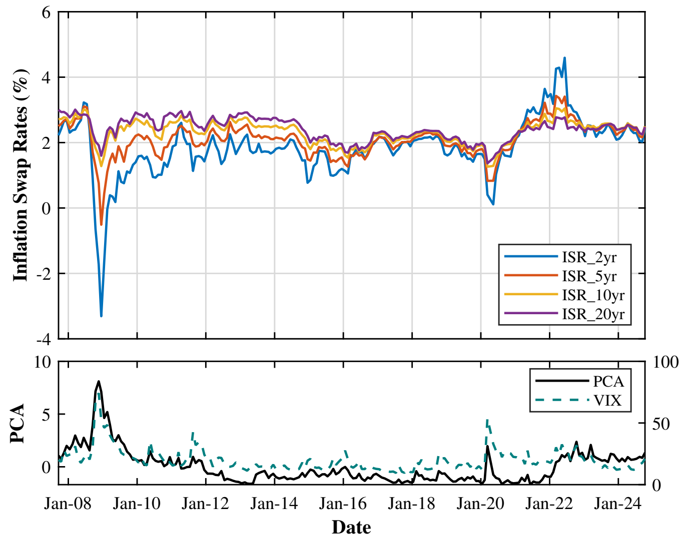
This assumption is validated following the same approach applied to λ_t^T:

- Visual inspection of the timeseries behaviour of inflation swap rates and the two liquidity proxies, VIX and PCA.
- Regression analysis of inflation swap rates and $\hat{\gamma}_t^T$ on VIX_t and PCA_t, respectively.

Results based on PCA and VIX are reported in Figure (5), Tables (6) and (7). Overall, inflation swap rates exhibit no statistically significant dependence on either liquidity measure across tenors. The R^2 values remain below 1%, and the p-values of the liquidity coefficients generally fail to reject the null hypothesis that $\hat{\beta} = 0$. Consistent with this, the estimated coefficients $\hat{\beta}_{VIX}$ are economically negligible and fluctuate around zero, taking both positive and negative values. Even when statistically significant, the magnitude of the effect remains small. For instance, a one-unit increase in VIX is associated with a decline in ISR^{20} of less than 2 basis points[39]. Hence, this evidence supports our assumption[40].

FIGURE V

Timeseries: ISR vs PCA and VIX[1]



[1]This figure presents the timeseries of inflation swap rates at 2-, 5-, 10-, and 20-year tenors, together with the first principal component (PCA) and VIX, over the sample period from August 2007 to September 2024. The first principal component is extracted from the variance-covariance matrix of: (i) VIX, (ii) the on-/off-the-run 10-year US Treasury spread, (iii) MOVE, and (iv) the average yield error between the observed Treasury yield curve and its fitted counterpart. Inflation swap rates are reported in percent, while the first principal component and VIX are expressed in standardized units.

[39]This finding is consistent with Haubrich et al. (2012). The authors report that the effects of liquidity crises on inflation swap pricing were largely *de minimis* during the global financial crisis, as dealers ceased hedging positions through TIPS and nominal Treasuries and instead operated primarily as inflation swap brokers.

[40]Results for the Breusch–Pagan and Wald tests for the respective regressions are not reported in Appendix A. Indeed, since the regression coefficients reported in Tables (6) and (7) are already not statistically different from zero, there is no need to compute SUR-adjusted standard errors for inference.

TABLE VI

Inflation Swap Rates and PCA: Regression Results[m]

Tenor	$\hat{\alpha}$	$\hat{\beta}_{PCA}$	$SE(\hat{\alpha})$	$SE(\hat{\beta}_{PCA})$	$t(\alpha)$	$p(\alpha)$	$t(\beta)$	$p(\beta)$	R^2
ISR^2	1.8515	-0.2107	0.0579	0.0362	31.987	0.0000	-5.814	0.0000	0.143
ISR^3	1.9548	-0.1338	0.0465	0.0291	42.069	0.0000	-4.600	0.0000	0.095
ISR^4	2.0361	-0.0854	0.0395	0.0247	51.543	0.0000	-3.454	0.0007	0.056
ISR^5	2.1041	-0.0575	0.0353	0.0221	59.682	0.0000	-2.603	0.0099	0.032
ISR^6	2.1603	-0.0398	0.0322	0.0202	67.010	0.0000	-1.972	0.0500	0.019
ISR^7	2.2112	-0.0179	0.0296	0.0185	74.716	0.0000	-0.968	0.3341	0.005
ISR^8	2.2508	-0.0066	0.0279	0.0175	80.588	0.0000	-0.376	0.7070	0.001
ISR^9	2.2878	0.0045	0.0265	0.0166	86.440	0.0000	0.270	0.7877	0.000
ISR^{10}	2.3243	0.0146	0.0250	0.0157	92.919	0.0000	0.932	0.3524	0.004
ISR^{12}	2.3626	0.0151	0.0242	0.0152	97.462	0.0000	0.992	0.3225	0.005
ISR^{15}	2.3927	0.0187	0.0237	0.0148	101.060	0.0000	1.264	0.2076	0.008
ISR^{20}	2.4113	0.0238	0.0240	0.0150	100.360	0.0000	1.580	0.1157	0.012

[m]This table reports OLS estimates of ISR_t^T regressed on the first principal component (PCA_t), along with corresponding inferential statistics, including R^2, t-statistics, and p-values. T-statistics and R^2 are rounded to three decimal places, while all other metrics are reported to four decimal places. The first principal component is extracted from the variance-covariance matrix of: (i) VIX, (ii) the on-/off-the-run 10-year US Treasury spread, (iii) MOVE, and (iv) the average yield error between the observed Treasury yield curve and its fitted counterpart. The regression analysis covers the sample period from August 2007 to September 2024.

TABLE VII

Inflation Swap Rates and VIX: Regression Results[n]

Tenor	$\hat{\alpha}$	$\hat{\beta}_{VIX}$	$SE(\hat{\alpha})$	$SE(\hat{\beta}_{VIX})$	$t(\alpha)$	$p(\alpha)$	$t(\beta)$	$p(\beta)$	R^2
ISR^2	1.8404	0.0104	0.1694	0.0093	10.865	0.0000	1.120	0.2645	0.008
ISR^3	1.9729	0.0052	0.1423	0.0078	13.867	0.0000	0.668	0.5054	0.003
ISR^4	2.0555	0.0027	0.1248	0.0068	16.475	0.0000	0.392	0.6953	0.001
ISR^5	2.1385	0.0001	0.1144	0.0063	18.699	0.0000	0.014	0.9891	0.000
ISR^6	2.2056	-0.0019	0.1072	0.0059	20.570	0.0000	-0.327	0.7444	0.001
ISR^7	2.2482	-0.0027	0.1014	0.0056	22.166	0.0000	-0.493	0.6230	0.002
ISR^8	2.2944	-0.0040	0.0964	0.0053	23.800	0.0000	-0.759	0.4491	0.004
ISR^9	2.3384	-0.0051	0.0919	0.0050	25.444	0.0000	-1.017	0.3107	0.007
ISR^{10}	2.3815	-0.0061	0.0874	0.0048	27.259	0.0000	-1.269	0.2065	0.011
ISR^{12}	2.4612	-0.0090	0.0830	0.0045	29.656	0.0000	-1.979	0.0497	0.025
ISR^{15}	2.4985	-0.0100	0.0796	0.0044	31.380	0.0000	-2.295	0.0231	0.034
ISR^{20}	2.5462	-0.0125	0.0782	0.0043	32.552	0.0000	-2.912	0.0041	0.054

[n]This table reports OLS estimates of ISR_t^T regressed on VIX_t, along with corresponding inferential statistics, including R^2, t-statistics, and p-values. T-statistics and R^2 are rounded to three decimal places, while all other metrics are reported to four decimal places. The regression analysis covers the sample period from January 2012 to September 2024. Compared with the previous table, the sample period for this regression has been shortened to avoid capturing the correlation between VIX and ISR driven by revisions in growth expectations during the global financial crisis. This choice is motivated by the view that there is no reason to expect investors to systematically prefer selling or buying inflation when market liquidity is poor.

To further assess the relationship between λ_t^T and liquidity, we perform the same regression analysis on $\hat{\gamma}_t^T$. As defined in Equation (6), $\hat{\gamma}_t^T = rp\pi_t^T - \lambda_t^T = BEI_t^T - \hat{\pi}_t^T$, and it can be interpreted as the difference between expected inflation under the risk-neutral measure $\mathbb{Q}$ (BEI_t^T) and the physical measure $\mathbb{P}$ ($\hat{\pi}_t^T$).

This decomposition is central to our identification strategy: if λ_t^T captures the liquidity component embedded in the breakeven inflation, then $\hat{\gamma}_t^T$ should reflect the same dependence only if inflation swap rates and, in particular, inflation risk premia do not exhibit a systematic dependence on liquidity proxies. Therefore, we test this implication to validate the initial hypothesis that inflation swap rates do not depend on liquidity risk. In particular, we expect to find a negative relationship between $\hat{\gamma}_t^T$ and the liquidity proxies, with a magnitude comparable to the one observed between λ_t^T

and VIX or λ_t^T and PCA. If so, we can conclude that $rp\pi_t^T$ does not have a meaningful relationship with liquidity and validate the *ISR/Liquidity* assumption.

As the reader can observe from Tables (8) and (9) below, results are consistent with our expectations.

TABLE VIII

$\hat{\gamma}$ and PCA: Regression Results[o]

Tenor	$\hat{\alpha}^T$	$\hat{\beta}^T_{PCA}$	SE($\hat{\alpha}^T$)	SE($\hat{\beta}^T_{PCA}$)	$t(\alpha)$	$p(\alpha)$	$t(\beta)$	$p(\beta)$	R^2
$\hat{\gamma}^{2Y}$	-0.1739	-0.4005	0.0487	0.0305	-3.573	0.0004	-13.141	0.0000	0.458
$\hat{\gamma}^{3Y}$	-0.0593	-0.3530	0.0401	0.0251	-1.481	0.1403	-14.068	0.0000	0.492
$\hat{\gamma}^{4Y}$	0.0219	-0.3111	0.0350	0.0219	0.624	0.5332	-14.180	0.0000	0.496
$\hat{\gamma}^{5Y}$	0.0835	-0.2710	0.0319	0.0200	2.617	0.0095	-13.563	0.0000	0.474
$\hat{\gamma}^{6Y}$	0.1301	-0.2326	0.0300	0.0188	4.339	0.0000	-12.385	0.0000	0.429
$\hat{\gamma}^{7Y}$	0.1643	-0.1968	0.0289	0.0181	5.688	0.0000	-10.880	0.0000	0.367
$\hat{\gamma}^{8Y}$	0.1880	-0.1646	0.0283	0.0177	6.640	0.0000	-9.286	0.0000	0.297
$\hat{\gamma}^{9Y}$	0.2030	-0.1368	0.0281	0.0176	7.228	0.0000	-7.782	0.0000	0.229
$\hat{\gamma}^{10Y}$	0.2106	-0.1136	0.0280	0.0175	7.516	0.0000	-6.476	0.0000	0.171
$\hat{\gamma}^{12Y}$	0.1487	-0.0766	0.0277	0.0174	5.366	0.0000	-4.418	0.0000	0.087
$\hat{\gamma}^{20Y}$	0.1035	-0.0628	0.0267	0.0167	3.883	0.0001	-3.761	0.0002	0.065

[o]This table reports OLS estimates of $\hat{\gamma}_t^T$ regressed on the first principal component (PCA_t), along with corresponding inferential statistics, including R^2, t-statistics, and p-values. T-statistics and R^2 are rounded to three decimal places, while all other metrics are reported to four decimal places. The first principal component is extracted from the variance-covariance matrix of: (i) VIX, (ii) the on-/off-the-run 10-year US Treasury spread, (iii) MOVE, and (iv) the average yield error between the observed Treasury yield curve and its fitted counterpart. The regression analysis covers the sample period from August 2007 to September 2024.

TABLE IX

$\hat{\gamma}$ and VIX: Regression Results[p]

Tenor	$\hat{\alpha}^T$	$\hat{\beta}^T_{VIX}$	SE($\hat{\alpha}^T$)	SE($\hat{\beta}^T_{VIX}$)	$t(\alpha)$	$p(\alpha)$	$t(\beta)$	$p(\beta)$	R^2
$\hat{\gamma}^{2Y}$	0.9945	-0.0585	0.1303	0.0059	7.632	0.0000	-9.844	0.0000	0.322
$\hat{\gamma}^{3Y}$	0.9463	-0.0504	0.1102	0.0050	8.590	0.0000	-10.020	0.0000	0.330
$\hat{\gamma}^{4Y}$	0.8964	-0.0438	0.0972	0.0044	9.226	0.0000	-9.880	0.0000	0.324
$\hat{\gamma}^{5Y}$	0.8393	-0.0378	0.0878	0.0040	9.558	0.0000	-9.447	0.0000	0.304
$\hat{\gamma}^{6Y}$	0.7758	-0.0323	0.0810	0.0037	9.581	0.0000	-8.752	0.0000	0.273
$\hat{\gamma}^{7Y}$	0.7095	-0.0273	0.0761	0.0035	9.323	0.0000	-7.863	0.0000	0.233
$\hat{\gamma}^{8Y}$	0.6442	-0.0228	0.0728	0.0033	8.848	0.0000	-6.877	0.0000	0.188
$\hat{\gamma}^{9Y}$	0.5830	-0.0190	0.0707	0.0032	8.245	0.0000	-5.899	0.0000	0.146
$\hat{\gamma}^{10Y}$	0.5278	-0.0159	0.0694	0.0032	7.602	0.0000	-5.015	0.0000	0.110
$\hat{\gamma}^{12Y}$	0.3838	-0.0118	0.0670	0.0031	5.729	0.0000	-3.853	0.0002	0.068
$\hat{\gamma}^{20Y}$	0.3121	-0.0104	0.0640	0.0029	4.880	0.0000	-3.582	0.0004	0.059

[p]This table reports OLS estimates of $\hat{\gamma}^T_t$ regressed on VIX_t, along with corresponding inferential statistics, including R^2, t-statistics, and p-values. T-statistics and R^2 are rounded to three decimal places, while all other metrics are reported to four decimal places. The regression analysis covers the sample period from August 2007 to September 2024.

7.1.3 Overall Conclusions on Factors Affecting Real Yields

Consolidating all the evidence presented in Sections (7.1.1) and (7.1.2), we conclude that TIPS yields embed a significant dependence on liquidity, as reflected in the sensitivity of λ^T_t and $\hat{\gamma}^T_t$ to VIX and PCA. It follows that TIPS yields are higher when liquidity is lower, especially during periods of economic distress. Therefore, the additional compensation offered by TIPS does not represent an arbitrage opportunity, but rather reflects compensation for bearing liquidity risk. At the same time, there is also a dependence on inflation surprises, indicating that inflation shocks increase demand for TIPS, pushing up bond prices and compressing λ[41].

[41]TIPS yields may depend on factors beyond liquidity and inflation expectations. Dittmar et al. (2026) documents a significant relationship between TIPS pricing and US default risk, as measured by CDS spreads, implying that variations in TIPS yields can also reflect changes in perceived sovereign credit risk. Therefore, liquidity conditions, inflation-related shocks, and sovereign credit risk may all contribute to the observed dynamics of TIPS yields.

7.2 Decomposition of Nominal (and Real) Yields

As shown in Equation (8) of the *Methodology* section, yields are decomposed into expectations and risk premia, under the assumption that yield convexity for both bond yields and inflation swap rates is negligible. The proposed decomposition requires not only market-observable quantities (e.g. ISR_t^T or Y_t^T), but also:

- an estimate of expected inflation: $\hat{\pi}_t^T$, and
- a $\mathbb{P}$-measure expectation of nominal yields

These measures, presented in Sections (6.2.1) and (6.2.2), allow for the identification of all *unknown* variables in Equation (8) and close the decomposition of yields into inflation and real rate expectations, as well as the associated risk premia.

7.2.1 Decomposition of Nominal Yields

As the Federal Reserve began reporting the *Dot Plot* only in January 2012, the analysis is restricted to a subset of the available timeseries. Expected inflation, the inflation risk premium, and nominal Treasury yields are reported at a monthly frequency, whereas the decomposition of nominal and real risk premia[42] is available only at a quarterly frequency, in line with the *Dot Plot* reporting schedule.

Figures (6), (7), (8), and (9) present the cross-sectional decomposition at the 5-, 10-, 15-, and 20-year tenors[43] over the period from January 2012 to September 2024.

[42] It is worth noting that the nominal risk premium corresponds to the difference between nominal Treasury yields and $\frac{\mathbb{E}_t^{\mathbb{P}}\left[\int t^T R(s)ds\right]}{\tau}$; hence, it is independent of expected average inflation. By contrast, expected average inflation affects the decomposition of the nominal risk premium into inflation and real risk premia. Accordingly, the subfigures corresponding to the inflation and real risk premia report estimates under both measures of expected average inflation (Federal Reserve and econometric).

[43] Results for the 2-year tenor are not considered representative due to the material effect of second-order dynamics, such as the indexation lag, on short-term bonds, which prevents a clean decomposition of the corresponding yields.

Since expected average inflation is not directly observable, and therefore decomposition results are *uncertain*, we compare results for two alternative measures of expected average inflation: (i) Federal Reserve estimates and (ii) AR(1)-based estimates. As reported in Section (6.2.1), the discrepancy between these two measures declines with maturity; hence, risk premia estimates become more reliable at longer tenors, where they are also more economically relevant.

The inflation risk charts show that the corresponding risk premium did not increase significantly, despite mounting inflation pressures toward the end of the COVID-19 period. Investors' expectations remained anchored to the Federal Reserve target with no evidence of a loss of credibility. At the same time, the uncertainty surrounding inflation expectations affects the decomposition between inflation and real risk premia. In particular, Federal Reserve estimates imply a significant divergence between real (negative) and inflation (positive) risk premia, which appears less realistic than the outcome produced by the AR(1)-based measure during 2022.

In general, since market expectations of inflation are not directly observable, either measure could provide a valid approximation. However, Federal Reserve estimates are structurally lower than the proposed econometric measure across all tenors, and therefore imply negative real risk premia and higher (positive) inflation risk premia. While this may be plausible, it raises an important question: *why are Federal Reserve-based expectations associated with a higher inflation risk premium?* Furthermore, the Federal Reserve estimate of expected average inflation at the 20-year horizon appears relatively volatile, in contrast to the common view that long-term expectations should be more stable, as suggested instead by the AR(1)-based measure.

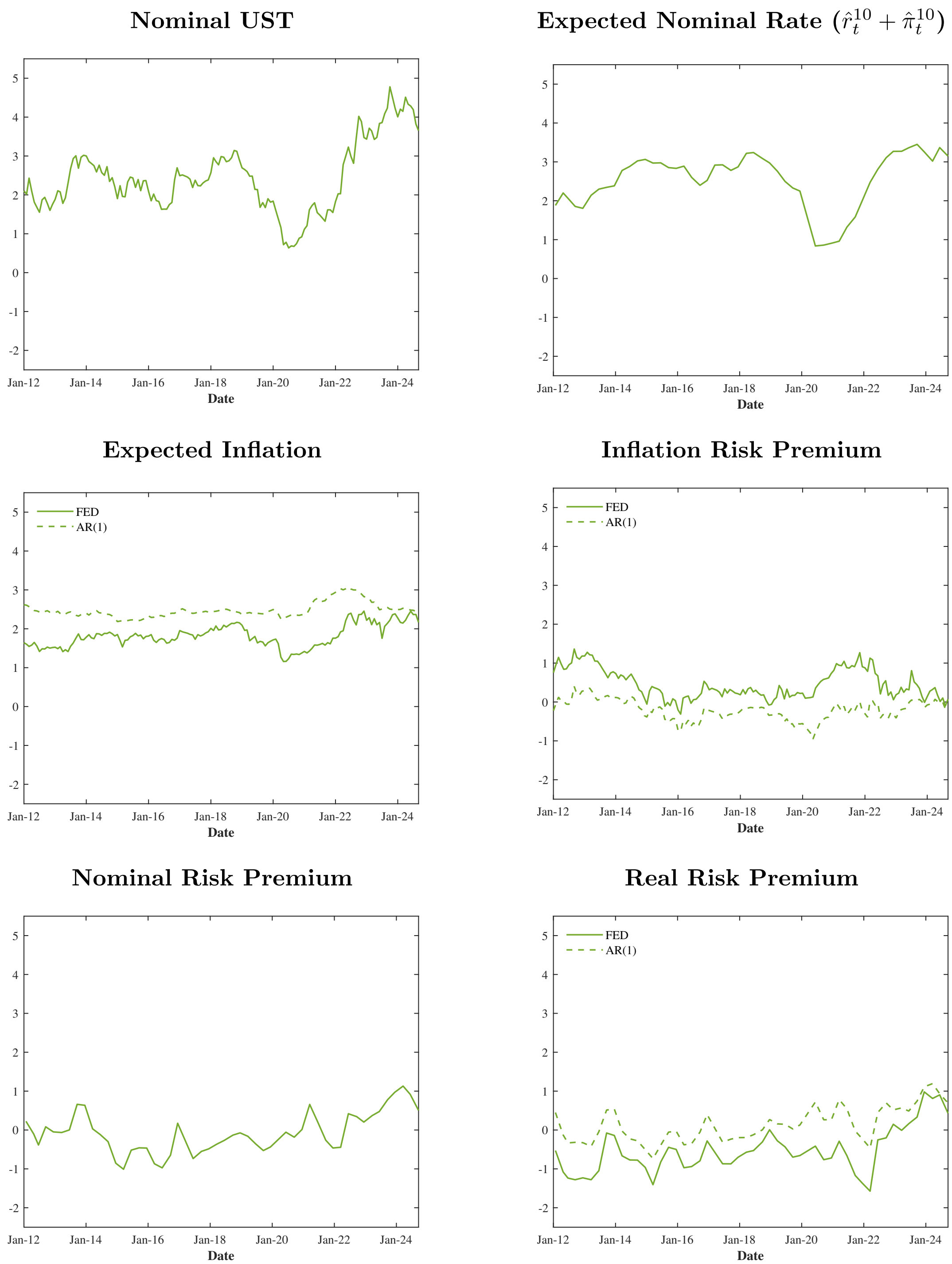
FIGURE VI

Nominal Yield System Identification: 5-Year Tenor[q]

[q]This figure reports six charts showing the Federal Reserve and AR(1) inflation timeseries, the inflation risk premium, the expected nominal rate, the nominal Treasury yield, the nominal risk premium, and the real risk premium at the 5-year tenor. The y-axis reports all metrics in % and the x-axis shows the corresponding date. The reported time period spans over January 2012 and September 2024.

FIGURE VII

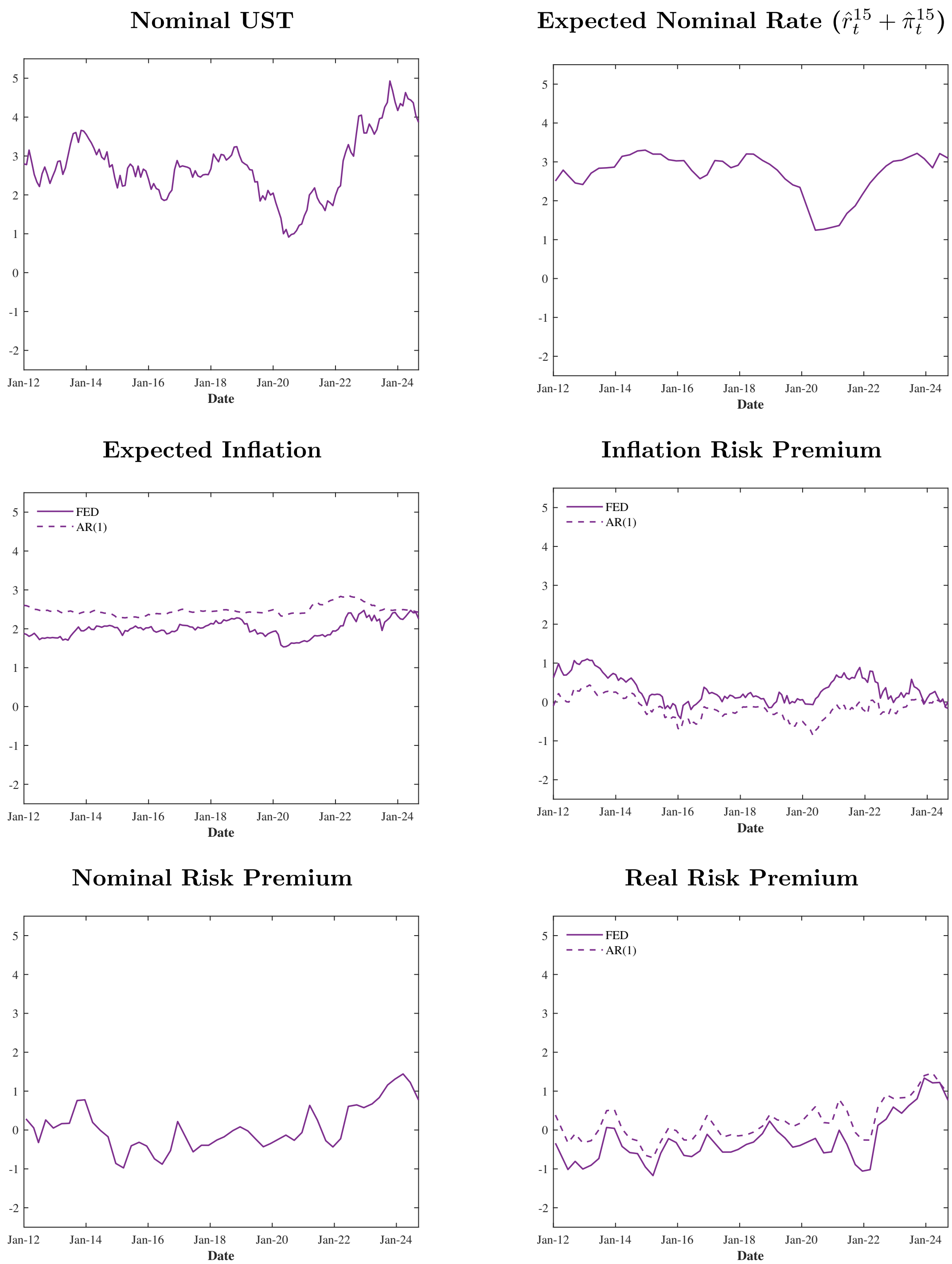
Nominal Yield System Identification: 10-Year Tenor[r]



[r]This figure reports six charts showing the Federal Reserve and AR(1) inflation timeseries, the inflation risk premium, the expected nominal rate, the nominal Treasury yield, the nominal risk premium, and the real risk premium at the 10-year tenor. The y-axis reports all metrics in % and the x-axis shows the corresponding date. The reported time period spans over January 2012 and September 2024.

FIGURE VIII

Nominal Yield System Identification: 15-Year Tenor[s]



[s]This figure reports six charts showing the Federal Reserve and AR(1) inflation timeseries, the inflation risk premium, the expected nominal rate, the nominal Treasury yield, the nominal risk premium, and the real risk premium at the 15-year tenor. The y-axis reports all metrics in % and the x-axis shows the corresponding date. The reported time period spans over January 2012 and September 2024.

FIGURE IX

Nominal Yield System Identification: 20-Year Tenor[t]

[t]This figure reports six charts showing the Federal Reserve and AR(1) inflation timeseries, the inflation risk premium, the expected nominal rate, the nominal Treasury yield, the nominal risk premium, and the real risk premium at the 20-year tenor. The y-axis reports all metrics in % and the x-axis shows the corresponding date. The reported time period spans over January 2012 and September 2024.

7.2.2 Decomposition of Real Yields

In the same spirit of the previous section, we provide the decomposition of $y_t^T = \hat{r}_t^T + rpr_t^T + \lambda_t^T$. By assessing these components, we focus on: (i) the effects of liquidity and inflation surprises (λ) and (ii) expectations and risk premia for real rates ($\hat{r} + rpr$) only, aiming to observe how the real risk premium performed during periods of heightened inflation (e.g. in 2022).

Figures (10), (11), (12), and (13) present the cross-sectional decomposition of TIPS yields at the 5-, 10-, 15-, and 20-year tenors[44] over the period from January 2012 to September 2024[45].

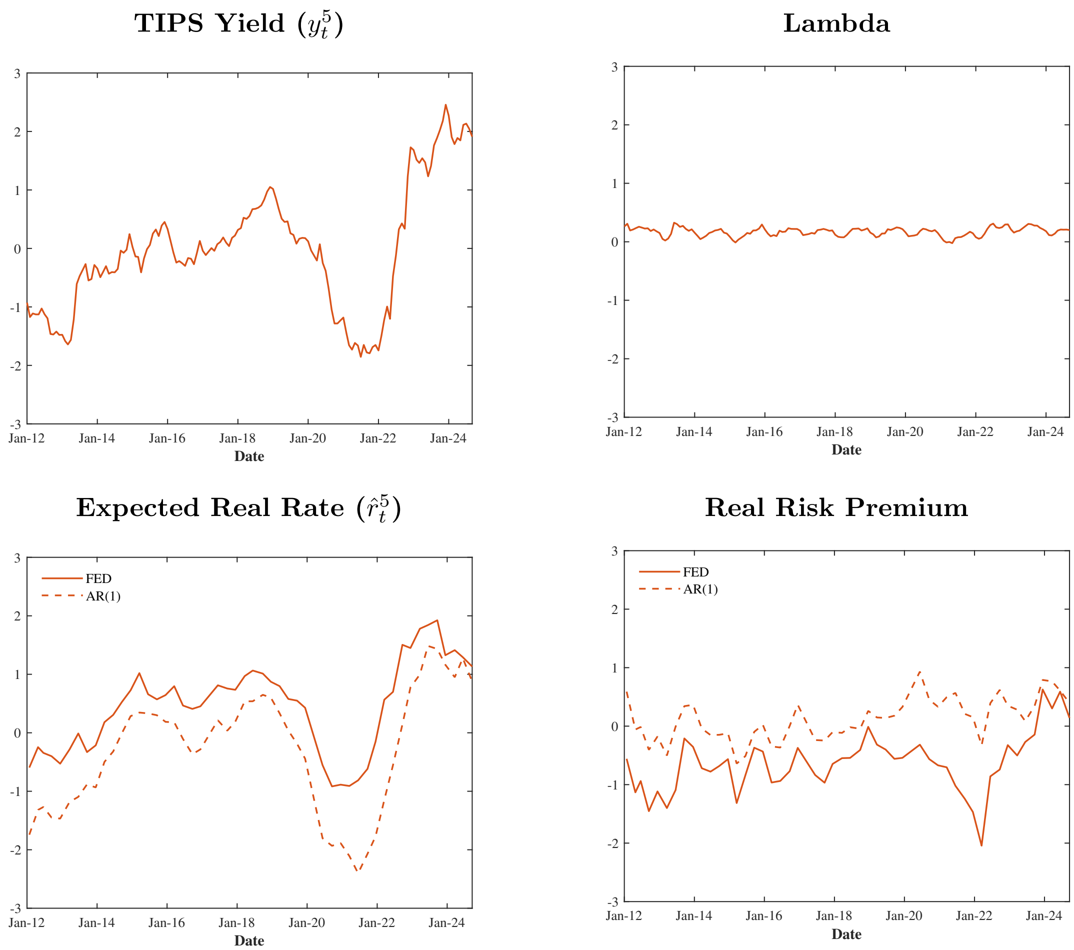
As an additional remark building on the findings from the decomposition of nominal yields, it is worth noting that real risk premia tend to become negative in 2022, at the onset of the post-COVID inflation surge. This could be potentially justified by investors placing greater value on the inflation protection provided by TIPS, thereby bidding up market prices and compressing real risk premia.

[44] Results for the 2-year tenor are not considered representative due to the material effect of second-order dynamics, such as the indexation lag, on short-term bonds, which prevents a clean decomposition of the corresponding yields.

[45] The expected real rate is obtained by subtracting expected average inflation from the expected nominal rate, described in Section (6.2.2).

FIGURE X

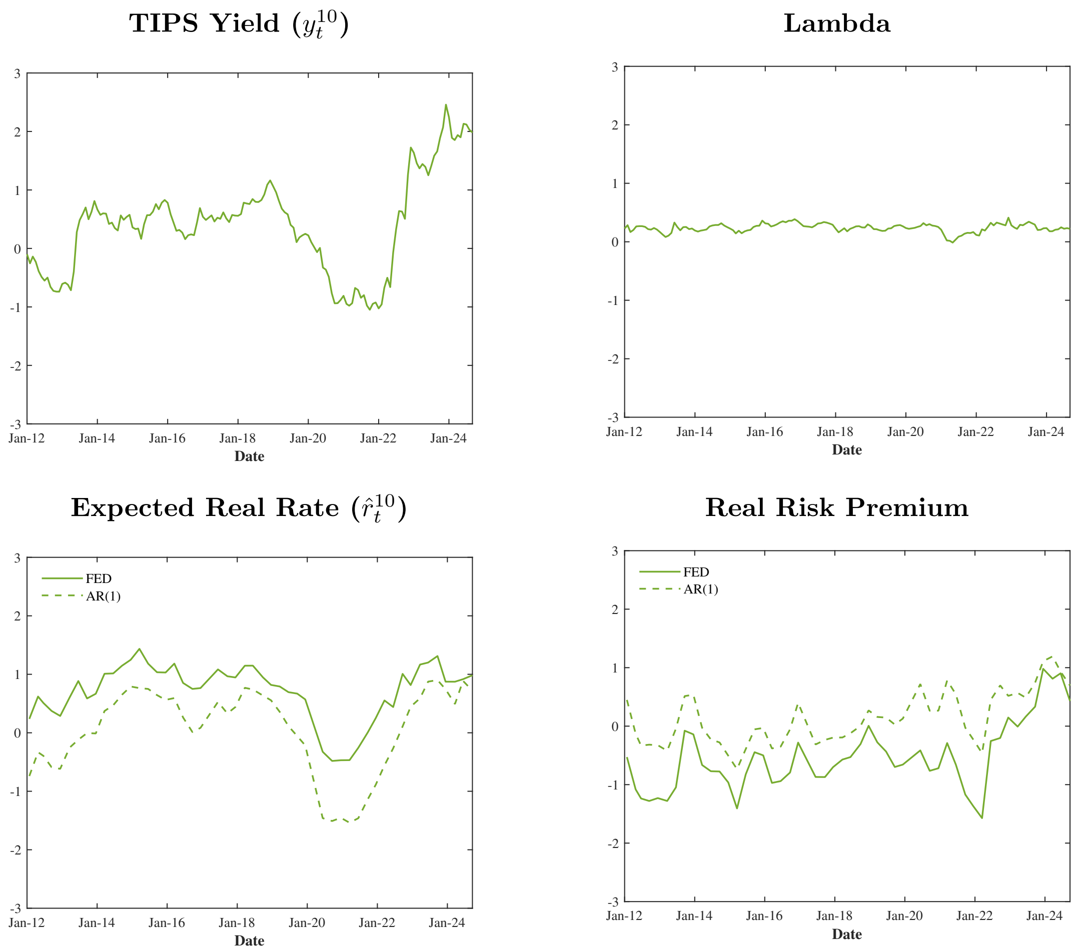
TIPS Yield System Identification: 5-Year Tenor[u]



[u]This figure reports four charts showing decomposition of TIPS yields at the 5-year tenor into expected real rate, real risk premium and λ. The y-axis reports all metrics in % and the x-axis shows the corresponding date. The reported time period spans over January 2012 and September 2024.

FIGURE XI

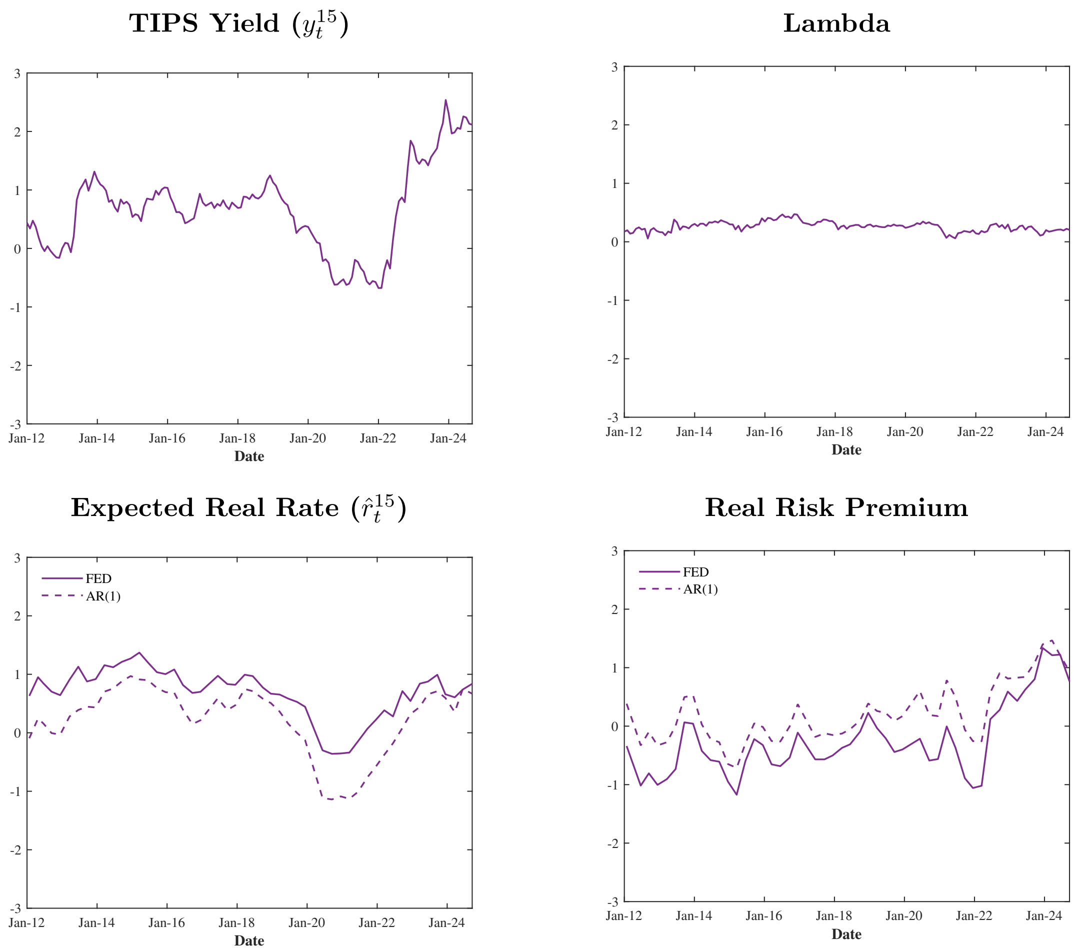
TIPS Yield System Identification: 10-Year Tenor[v]



[v]This figure reports four charts showing decomposition of TIPS yields at the 10-year tenor into expected real rate, real risk premium and λ. The y-axis reports all metrics in % and the x-axis shows the corresponding date. The reported time period spans over January 2012 and September 2024.

FIGURE XII

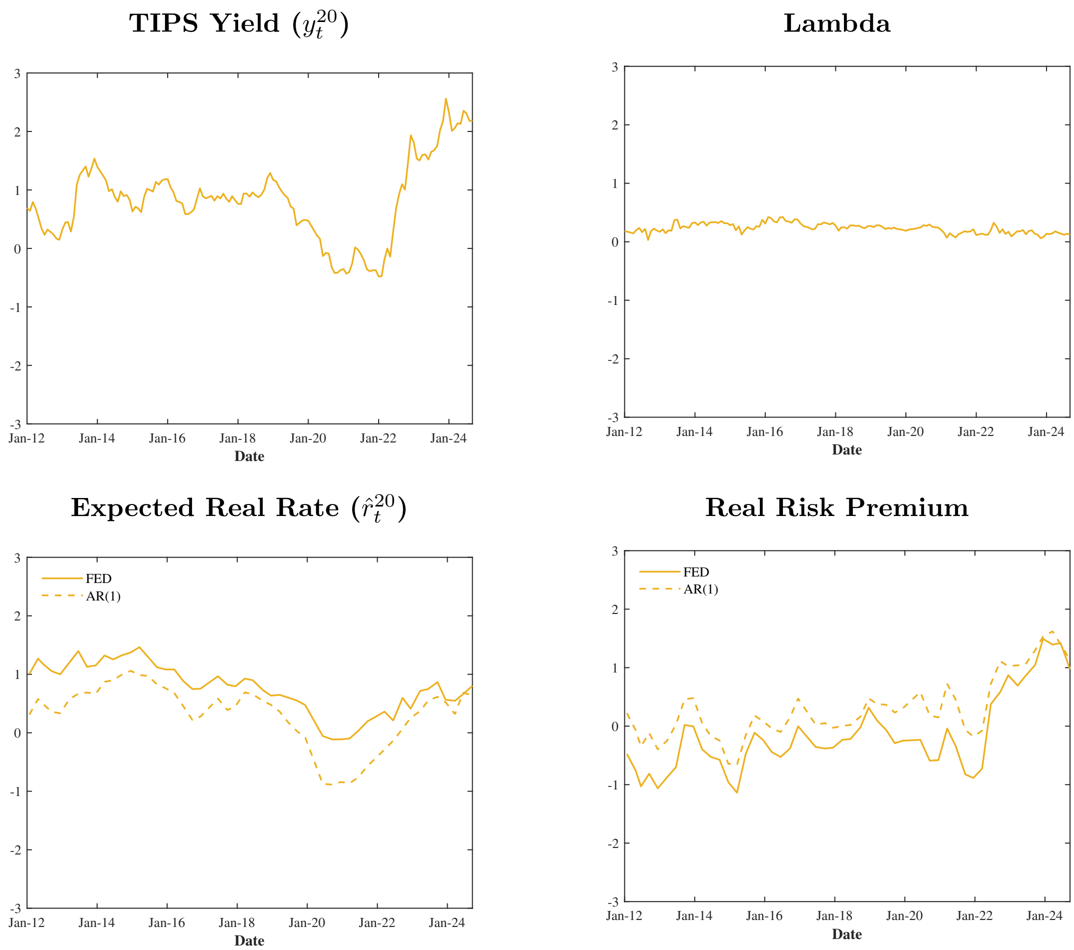
TIPS Yield System Identification: 15-Year Tenor[z]



[z]This figure reports four charts showing decomposition of TIPS yields at the 15-year tenor into expected real rate, real risk premium and λ. The y-axis reports all metrics in % and the x-axis shows the corresponding date. The reported time period spans over January 2012 and September 2024.

FIGURE XIII

TIPS Yield System Identification: 20-Year Tenor[aa]



[aa]This figure reports four charts showing decomposition of TIPS yields at the 20-year tenor into expected real rate, real risk premium and λ. The y-axis reports all metrics in % and the x-axis shows the corresponding date. The reported time period spans over January 2012 and September 2024.

8 Conclusions

This paper investigated whether inflation expectations remained anchored and if TIPS were subject to arbitrage opportunities by employing a linear yield decomposition framework for TIPS, nominal US Treasuries, and inflation swap rates.

While the methodology relies on a few simplifying assumptions, its main strength consists in allowing both the timeseries and cross-sectional analysis of bond yields, and the robustness of our findings has been tested against multiple proxies for the variables employed (liquidity and expected inflation).

Our findings indicate that TIPS are not mispriced in comparison with nominal Treasuries. Instead, short/medium term TIPS yields embed an additional premium reflecting co-movement with liquidity risk and long term TIPS are sensitive to inflation surprises: real risk premia become negative during phases of high inflation. This is at first blush surprising, because, apart from minor indexation effects, the price of TIPS is derived by discounting its real cashflows at the real rate. We conjecture that this phenomenon could be explained by market segmentation and leave empirical testing of this conclusion as a next step for further research.

As an additional result, inflation expectations remained broadly anchored to the Federal Reserve target after the COVID-19 inflation surge. Nevertheless, Federal Reserve estimates appeared lower and converged more rapidly to the long-term target than the proposed econometric measure, resulting in higher inflation risk premia. This pattern may reflect the optimism of the Federal Reserve in keeping inflation under control.

9 Appendix A

The objective of this appendix is twofold: (i) to present the framework of an additional econometric tool complementing the OLS regression analysis previously applied, and (ii) to elaborate further results supporting the validity of our assumptions and quantifying the strength of the various dependencies among relevant quantities, such as λ_t^T, $\hat{\gamma}_t^T$, and ISR_t^T, on liquidity proxies, VIX_t and PCA_t.

9.1 SUR Framework

Since it is sensible to assume that: (i) the dependent variables are connected via cross-correlated shocks[46] and (ii) the idiosyncratic shocks do not have the same intensity across different tenors (heteroskedasticity), we propose a seemingly unrelated regressions (SUR) model as an additional tool supporting our analysis. Although the regressors are identical across the respective equations encompassing the full maturity range, and thus there is no difference in coefficient estimates relative to OLS, it is worth noting that this model provides the appropriate covariance structure of the coefficients for conducting joint tests on the estimates (e.g. Wald tests).

The model is presented for λ_t^T and $\hat{\gamma}_t^T$ as follows[47]:

$$\begin{bmatrix} \boldsymbol{\lambda^2} \\ \boldsymbol{\lambda^3} \\ \vdots \\ \boldsymbol{\lambda^{20}} \end{bmatrix} = \begin{bmatrix} \boldsymbol{X_\lambda^2} & 0 & \cdots & 0 \\ 0 & \boldsymbol{X_\lambda^3} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \boldsymbol{X_\lambda^{20}} \end{bmatrix} \begin{bmatrix} \boldsymbol{\beta_\lambda^2} \\ \boldsymbol{\beta_\lambda^3} \\ \vdots \\ \boldsymbol{\beta_\lambda^{20}} \end{bmatrix} + \begin{bmatrix} \boldsymbol{u_\lambda^2} \\ \boldsymbol{u_\lambda^3} \\ \vdots \\ \boldsymbol{u_\lambda^{20}} \end{bmatrix} \tag{12}$$

[46] As an example, monetary policy interventions which simultaneously impact the term structure of the BEI.

[47] Although we test the conjecture that inflation swap rates are not affected by liquidity by estimating a SUR model between ISR_t^T and the distinct liquidity proxies, we do not repeat the model specification for this variable, because there are no changes in the setup and to preserve clarity of presentation.

$$\begin{bmatrix} \boldsymbol{\hat{\gamma}^2} \\ \boldsymbol{\hat{\gamma}^3} \\ \vdots \\ \boldsymbol{\hat{\gamma}^{20}} \end{bmatrix} = \begin{bmatrix} \boldsymbol{X^2_{\hat{\gamma}}} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \boldsymbol{X^3_{\hat{\gamma}}} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \boldsymbol{X^{20}_{\hat{\gamma}}} \end{bmatrix} \begin{bmatrix} \boldsymbol{\beta^2_{\hat{\gamma}}} \\ \boldsymbol{\beta^3_{\hat{\gamma}}} \\ \vdots \\ \boldsymbol{\beta^{20}_{\hat{\gamma}}} \end{bmatrix} + \begin{bmatrix} \boldsymbol{u^2_{\hat{\gamma}}} \\ \boldsymbol{u^3_{\hat{\gamma}}} \\ \vdots \\ \boldsymbol{u^{20}_{\hat{\gamma}}} \end{bmatrix} \tag{13}$$

The defintions and structure behind Equations (12) and (13) are presented below for $\hat{\gamma}_t^T$[48]:

$$\hat{\gamma}_t^T = \alpha^T + \beta_{\hat{\gamma}}^T PCA_t + u_t^T$$

$$\boldsymbol{\hat{\gamma}_t} = \boldsymbol{\alpha} + \boldsymbol{\beta_{\hat{\gamma}}} PCA_t + \boldsymbol{u_t}$$

where:

- $\boldsymbol{\hat{\gamma}_t} = \begin{bmatrix} \hat{\gamma}_t^2 & \ldots & \hat{\gamma}_t^{10} & \hat{\gamma}_t^{12} & \hat{\gamma}_t^{15} & \hat{\gamma}_t^{20} \end{bmatrix}'$. It is a vector (12 x 1) that contains $\hat{\gamma}_t^T$ for $T = 2, \ldots, 10, 12, 15, 20$.

- $\boldsymbol{\alpha} = \begin{bmatrix} \alpha^2 & \ldots & \alpha^{10} & \alpha^{12} & \alpha^{15} & \alpha^{20} \end{bmatrix}'$ and represents the vector of the intercepts of dimension 12x1.

- $\boldsymbol{\beta_{\hat{\gamma}}} = \begin{bmatrix} \beta_{\hat{\gamma}}^2 & \ldots & \beta_{\hat{\gamma}}^{10} & \beta_{\hat{\gamma}}^{12} & \beta_{\hat{\gamma}}^{15} & \beta_{\hat{\gamma}}^{20} \end{bmatrix}'$. It is the vector (12 x 1) of the coefficients on PCA.

- $\boldsymbol{u_t} = \begin{bmatrix} u_t^2 & \ldots & u_t^{10} & u_t^{12} & u_t^{15} & u_t^{20} \end{bmatrix}'$, representing the vector of the regression errors with dimension 12 x 1.

Stacking the 12 systems of equations[49] into one matrix expression yields a more compact

[48] The same framework applies to λ_t^T. Furthermore, the regressor used here is PCA. The same definitions and vectors apply for VIX as well.

[49] This is the overall number of different tenors being assessed, with each having 204 monthly observations covering the period from August 2007 up to September 2024. The dataset contains a

formula which will be useful for the estimation of the model:

$$\underbrace{\begin{bmatrix} \hat{\boldsymbol{\gamma}}^{\mathbf{2}} \\ \hat{\boldsymbol{\gamma}}^{\mathbf{3}} \\ \vdots \\ \hat{\boldsymbol{\gamma}}^{\mathbf{20}} \end{bmatrix}}_{\substack{=\hat{\boldsymbol{\gamma}} \\ (2448\times 1)}} = \underbrace{\begin{bmatrix} \boldsymbol{X}^{\mathbf{2}}_{\hat{\gamma}} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \boldsymbol{X}^{\mathbf{3}}_{\hat{\gamma}} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \boldsymbol{X}^{\mathbf{20}}_{\hat{\gamma}} \end{bmatrix}}_{\substack{=\boldsymbol{X}^{*}_{\hat{\gamma}} \\ (2448\times 24)}} \underbrace{\begin{bmatrix} \boldsymbol{\beta}^{\mathbf{2}}_{\hat{\gamma}} \\ \boldsymbol{\beta}^{\mathbf{3}}_{\hat{\gamma}} \\ \vdots \\ \boldsymbol{\beta}^{\mathbf{20}}_{\hat{\gamma}} \end{bmatrix}}_{\substack{=\boldsymbol{\beta}_{\hat{\gamma}} \\ (24\times 1)}} + \underbrace{\begin{bmatrix} \boldsymbol{u}^{\mathbf{2}}_{\hat{\gamma}} \\ \boldsymbol{u}^{\mathbf{3}}_{\hat{\gamma}} \\ \vdots \\ \boldsymbol{u}^{\mathbf{20}}_{\hat{\gamma}} \end{bmatrix}}_{\substack{=\boldsymbol{u}_{\hat{\gamma}} \\ (2448\times 1)}}$$

where:

- $\hat{\boldsymbol{\gamma}}$ is the vector stacking the timeseries observations per each $\hat{\boldsymbol{\gamma}}^{\boldsymbol{T}}$ for $T = 2, \ldots, 12, 15, 20$.
- $\boldsymbol{X}^{*}_{\hat{\gamma}}$ is the matrix of the regressors, reporting on its diagonal each $\boldsymbol{X}^{\boldsymbol{T}}_{\hat{\gamma}}$ for $T = 2, \ldots, 12, 15, 20$.
- $\boldsymbol{X}^{\boldsymbol{T}}_{\hat{\gamma}} = \underbrace{\begin{bmatrix} 1 & PCA_{Aug_{07}} \\ 1 & PCA_{Sep_{07}} \\ \vdots & \vdots \\ 1 & PCA_{Sep_{24}} \end{bmatrix}}_{\substack{=\boldsymbol{X}^{T} \\ (204\times 2)}}$
- $\boldsymbol{\beta}_{\hat{\gamma}}$ is the vector grouping the regression coefficients such that:

$$\boldsymbol{\beta}^{\boldsymbol{T}}_{\hat{\gamma}} = \underbrace{\begin{bmatrix} \alpha^{T} & \beta^{T}_{\hat{\gamma}} \end{bmatrix}'}_{\substack{=\boldsymbol{\beta}^{\boldsymbol{T}}_{\hat{\gamma}} \\ (2\times 1)}}$$

- $\boldsymbol{u}_{\hat{\gamma}}$ is the vector stacking the timeseries observations per each $\boldsymbol{u}^{\boldsymbol{T}}_{\hat{\gamma}}$ for $T = 2, \ldots, 12, 15, 20$.

total of 2448 observations.

The system can be estimated by OLS under standard assumptions for T fixed and $t \to \infty$ and the corresponding estimator is[50]:

$$\hat{\boldsymbol{\beta}}_{\hat{\gamma}} = \left(\boldsymbol{X}^{*\prime}_{\hat{\gamma}} \boldsymbol{X}^{*}_{\hat{\gamma}}\right)^{-1} \boldsymbol{X}^{*\prime}_{\hat{\gamma}} \hat{\gamma}$$

$$= \begin{bmatrix} \left(\boldsymbol{X}^{\boldsymbol{2}\prime}_{\hat{\gamma}} \boldsymbol{X}^{\boldsymbol{2}}_{\hat{\gamma}}\right)^{-1} \boldsymbol{X}^{\boldsymbol{2}\prime}_{\hat{\gamma}} \hat{\gamma}^{\boldsymbol{2}} \\ \vdots \\ \left(\boldsymbol{X}^{\boldsymbol{20}\prime}_{\hat{\gamma}} \boldsymbol{X}^{\boldsymbol{20}}_{\hat{\gamma}}\right)^{-1} \boldsymbol{X}^{\boldsymbol{20}\prime}_{\hat{\gamma}} \hat{\gamma}^{\boldsymbol{20}} \end{bmatrix}$$

To conclude, the variance-covariance matrix of $\hat{\boldsymbol{\beta}}$ is defined via the classic sandwich structure:

$$V(\hat{\boldsymbol{\beta}}_{\hat{\gamma}}|\boldsymbol{X}^{*}_{\hat{\gamma}}) = \left(\boldsymbol{X}^{*\prime}_{\hat{\gamma}} \boldsymbol{X}^{*}_{\hat{\gamma}}\right)^{-1} \boldsymbol{X}^{*\prime}_{\hat{\gamma}} \left(\boldsymbol{\Omega}_{\hat{\gamma}} \otimes \mathbb{I}_{\boldsymbol{204}}\right) \boldsymbol{X}^{*}_{\hat{\gamma}} \left(\boldsymbol{X}^{*\prime}_{\hat{\gamma}} \boldsymbol{X}^{*}_{\hat{\gamma}}\right)^{-1}$$

with:

- $\mathbb{I}_{\boldsymbol{204}}$ represents the identity matrix of dimension 204×204.
- $\boldsymbol{\Omega}_{\hat{\gamma}}$ is the contemporaneous variance-covariance matrix of the residuals, which is assumed finite and positive-definite[51]:

$$V(\boldsymbol{u_t}) = \boldsymbol{\Omega}_{\hat{\gamma}} = \begin{bmatrix} w_2^2 & w_{2,3} & \cdots & w_{2,20} \\ \vdots & \vdots & \ddots & \vdots \\ w_{20,2} & w_{20,3} & \cdots & w_{20}^2 \end{bmatrix}$$

Making use of OLS residuals, each element $w_{T,J} \quad \forall\, T, J$ can be estimated using

[50] Each $\boldsymbol{\beta}^{\boldsymbol{T}}_{\hat{\gamma}}$ can be estimated using OLS equation-by-equation across 12 different samples consisting of 204 observations each.

[51] For ease of reading, the tenor suffix is applied as a pedix in this instance.

the following expression[52]:

$$w_{T,J} = \frac{1}{204} \sum_{t=1}^{204} \hat{w}_{T,t} \hat{w}_{J,t}$$

9.2 SUR Results

The second part of this appendix expands on the analysis of the residuals obtained in the OLS regressions and assesses whether:

- the residuals are cross-correlated by performing a Breusch-Pagan test, hence validating whether a SUR framework is needed,
- the regression coefficients are jointly equal to zero via a Wald test, adjusting for the appropriate covariance structure of the residuals.

The results presented in Tables (10), (11), (12), (13), and (14), covering the respective regressions of λ and $\hat{\gamma}$ on the separate liquidity proxies and of λ on $InfS$, show that, although the residuals tend to co-move across tenors, the regression coefficients remain statistically different from zero even after accounting for a non-diagonal error correlation matrix.

[52] $w_{T,T} = w_T^2$.

TABLE A.1

λ Regressions: Breusch-Pagan Tests[A.a]

Regression	BP LM χ^2	df	p-value
VIX	6036.533	55	0.0000
PCA	5977.250	55	0.0000

[A.a]This table presents the test statistics for the null hypothesis that the OLS residuals, obtained from the regressions of λ_t^T on PCA_t and VIX_t, respectively, are uncorrelated across tenor equations. The first principal component is extracted from the variance-covariance matrix of: (i) VIX, (ii) the on-/off-the-run 10-year US Treasury spread, (iii) MOVE, and (iv) the average yield error between the observed Treasury yield curve and its fitted counterpart. The Breusch-Pagan statistic is computed from pairwise residual correlations. Rejection of the null indicates significant cross-sectional correlation, motivating the use of a SUR covariance structure.

TABLE A.2

λ Regressions: Wald Tests[A.b]

Regression	Wald χ^2	df	p-value
VIX	359.645	11	0.0000
PCA	754.285	11	0.0000

[A.b]The table reports the Wald χ^2 statistic and associated p-value for the null hypothesis that all slope coefficients on the explanatory variable are jointly equal to zero across maturities, $H_0 : \hat{\beta}^2 = \cdots = \hat{\beta}^{20} = 0$. The test is conducted within a SUR framework, thereby accounting for cross-equation correlation in the residuals. Results are reported for alternative specifications using VIX_t and PCA_t as regressors. The first principal component is extracted from the variance-covariance matrix of: (i) VIX, (ii) the on-/off-the-run 10-year US Treasury spread, (iii) MOVE, and (iv) the average yield error between the observed Treasury yield curve and its fitted counterpart.

TABLE A.3

$\hat{\gamma}$ Regressions: Breusch-Pagan Tests[A.c]

Regressor	Statistic	p-value
PCA	8952.050	<0.001
VIX	8908.200	<0.001

[A.c]This table presents the test statistics for the null hypothesis that the OLS residuals, obtained from the regressions of $\hat{\gamma}_t^T$ on PCA_t and VIX_t, respectively, are uncorrelated across tenor equations. The first principal component is extracted from the variance-covariance matrix of: (i) VIX, (ii) the on-/off-the-run 10-year US Treasury spread, (iii) MOVE, and (iv) the average yield error between the observed Treasury yield curve and its fitted counterpart. The regression analysis covers the sample period from August 2007 to September 2024. The Breusch-Pagan statistic is computed from pairwise residual correlations. Rejection of the null indicates significant cross-sectional correlation, motivating the use of a SUR covariance structure.

TABLE A.4

$\hat{\gamma}$ Regressions: Wald Tests[A.d]

Regressor	Statistic	p-value
PCA	775.011	<0.001
VIX	197.997	<0.001

[A.d]The table reports the Wald χ^2 statistic and associated p-value for the null hypothesis that all slope coefficients on the explanatory variable are jointly equal to zero across maturities, $H_0 : \hat{\beta}^2 = \cdots = \hat{\beta}^{20} = 0$. The test is conducted within a SUR framework, thereby accounting for cross-equation correlation in the residuals. Results are reported for alternative specifications using VIX_t and PCA_t as regressors. The first principal component is extracted from the variance-covariance matrix of: (i) VIX, (ii) the on-/off-the-run 10-year US Treasury spread, (iii) MOVE, and (iv) the average yield error between the observed Treasury yield curve and its fitted counterpart. The regression analysis covers the sample period from August 2007 to September 2024.

TABLE A.5

λ Regressions vs $InfS$: Breusch-Pagan and Wald Tests[A.e]

Regressor	Statistic	p-value
Breusch–Pagan LM	7013.612	<0.001
Wald Test	72.094	<0.001

[A.e]The table reports the Breusch-Pagan and the Wald χ^2 statistic and associated p-value for the respective null hypothesis that: (i) the OLS residuals, obtained from the regressions of λ_t^T on $InfS$ are uncorrelated across tenor equations and (ii) all slope coefficients on the explanatory variable are jointly equal to zero across maturities, $H_0 : \hat{\beta}^2 = \cdots = \hat{\beta}^{20} = 0$. The Wald test is conducted within a SUR framework, thereby accounting for cross-equation correlation in the residuals. Results are reported for $InfS_t$ as regressor. The regression analysis covers the sample period from August 2007 to September 2024.

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